Banking Fragility and the Capitalization Dilemma*

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Abstract

Bank capital plays a dual role: abundant capital relaxes banks' balance-sheet constraints and expands the scale of credit, amplifying productivity shocks, while scarce capital leaves the banking system endogenously fragile and crisis-prone. We embed both channels in a single general-equilibrium model and analyze it globally. Their interaction produces a capitalization dilemma: the effect of bank capital on macroeconomic stability is non-monotonic, a feature invisible to single-channel models. A social planner who internalizes the fragility externality holds more capital, sharply cutting the frequency and cost of financial crises while leaving the economy more sensitive to productivity shocks. We distill the planner's solution into a simple time-consistent Taylor-rule-type leverage rule: its slope prices the fragility externality and rewards bank capitalization, tightening when credit growth erodes the sector-wide capital ratio, while its level accommodates booms driven by strong fundamentals. The rule recovers almost all of the planner's welfare gain, and by reading bank capital and output as separate signals it distinguishes credit-driven from output-driven booms, a distinction the credit-to-GDP gap conflates.

Keywords: Banking fragility, bank capitalization, macroprudential policy, occasionally binding constraints, global nonlinear solution.

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* Any views expressed here are solely those of the authors and do not represent those of the Bank of England or any of its committees.

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1 Introduction

Bank capital requirements lie at the heart of macroeconomic and financial stability debates. Systemic buffers such as the Basel III countercyclical capital buffer are already part of the macroprudential toolkit, but their optimal level, timing and state-contingency remain contested. The policy puzzle is simple. Thinly capitalized banks are fragile: they are more exposed to failures, runs and funding stress when confidence falls. But well-capitalized banks can also expand credit more aggressively in good times, increasing the economy's exposure when adverse real shocks arrive. The banking turmoil of early 2023 (Gruenberg, 2024; Basel Committee on Banking Supervision, 2023) brought both forces into view. It showed that capital matters for resilience, but also that the design and communication of buffers remain central policy questions. How high should bank capital requirements be, and when should they tighten or relax, if the same buffer can reduce systemic fragility while changing the transmission of normal downturns?

Existing models study these two faces of bank capital largely in isolation: crisis-buffer models on one side, financial-accelerator models on the other. This paper fills that gap by studying both effects in one general-equilibrium model. The contributions of this paper are threefold. First, we show that combining a *fragility channel* (high leverage heightens run risk) with a *credit-exposure channel* (strong balance sheets amplify real shocks) gives rise to a fundamental *capitalization dilemma*. In our model, any fixed capital requirement involves a trade-off: higher bank capital significantly improves resilience to financial crises (reducing the frequency and severity of crises) but simultaneously increases banks' exposure to real-sector downturns (as they build larger balance sheets and intermediate more in good times). In other words, the relationship between bank capital and macroeconomic stability is *non-monotonic*—a novel insight that would be missed in models focusing on only one channel.

Second, we quantify the gap between privately optimal and socially optimal capital levels. A social planner internalizing the fragility externality would choose meaningfully higher bank capital buffers than the private competitive equilibrium. This extra capital leads to lower crisis incidence and reduces output losses during financial crises, yet it leaves the economy more sensitive to ordinary productivity-driven recessions. The planner's intervention offers no offsetting gain in purely real (non-financial) downturns, and if anything slightly amplifies them, so the gains from regulation come almost entirely from mitigating financial crises. Quantitatively, moving to the planner's allocation is worth a permanent 1.3% of consumption, measured inclusive of the transition to higher capital. The planner's allocation also weakly dominates every constant leverage cap that improves on the competitive equilibrium, simply because any such cap's allocation is feasible for the planner. Relative to the competitive equilibrium, the planner cuts crisis frequency roughly tenfold (from 8.9% to 0.9%) and *raises* average output rather than depressing it (both in the ergodic distribution). The gain is therefore not a trade of output for safety but an improvement on both margins at once.

The third contribution is policy implementation. We show that the planner's allocation is closely approximated by a simple, time-consistent Taylor-rule-type leverage rule with two coefficients: a slope on bank capital and a level.[1] The slope prices the fragility externality: after the leverage constraint's shadow value, it is the Pigouvian wedge that rewards capitalization, and it does the welfare work, with the rule recovering about 97% of the planner's gain. The level accommodates the business cycle (looser when output is high), a refinement that adds information for the regulator rather than further welfare. By conditioning on bank capital and output, both observable unlike the latent productivity shock, the rule distinguishes credit growth that erodes the sector-wide capital ratio from output-driven growth, a distinction that fixed capital ratios or credit-to-GDP gap triggers obscure.

Underpinning these contributions is a quantitative model that combines ingredients from two strands of literature. We incorporate an endogenous *fragility (run) risk* that increases when bank equity is scarce, as in the banking-crisis models of Gertler and Kiyotaki (2010) and others. At the same time, banks in our model face an occasionally binding leverage constraint, so that high net worth translates into a lending boom and stronger transmission of real shocks (as in Brunnermeier and Sannikov (2014) among others). By integrating these two channels within one framework, we capture interactions that would be missed if each mechanism were studied in isolation. This integration is crucial for understanding why bank capital requirements entail a trade-off: ample capital acts as insurance against funding crises, but also supports larger balance sheets and greater macroeconomic sensitivity.

These mechanisms are not merely theoretical. Drawing on the long-run cross-country panel of Jordà et al. (2017), Section 2 documents three patterns that motivate the model's two channels: thinly capitalized banking systems are more prone to distress; better-capitalized systems tend to expand credit more rapidly (a pattern robust in sign, though its significance is specification-sensitive); and economies hit by adverse shocks after a

[1] Time-consistency is inherited from the planner's Markov structure. The constrained planner solves a recursive problem in the current aggregate state $X = (N, A)$, so its optimal policy is a Markov allocation involving no promises about future actions: re-optimizing at any date reproduces the same policy. Because the leverage rule mimics this Markov allocation and conditions only on current observables, a regulator following it has no incentive to deviate ex post and requires no commitment device.

credit boom suffer deeper output losses. Each fact maps to one link in the dual role of bank capital (a buffer in rare crises, an amplifier of ordinary shocks) which the remainder of the paper formalizes, quantifies, and translates into a macroprudential rule.

Related literature. This paper connects several strands of work on financial intermediation, banking fragility, and macroprudential policy.

It first builds on models of *banking panics and fragility*. Self-fulfilling depositor runs date to Diamond and Dybvig (1983), with the run probability microfounded as a global-games equilibrium by Goldstein and Pauzner (2005). Quantitative macroeconomic models embed such panics in general equilibrium: Gertler et al. (2020) generate endogenous runs that amplify downturns, Boissay et al. (2016) derive banking crises endogenously from credit booms, and He and Xiong (2012) study rollover risk in intermediary funding. In closely related contemporaneous work, Amador and Bianchi (2026) microfound a similar externality in a model where banks default for fundamental or self-fulfilling reasons and over-borrow because individual leverage depresses the asset prices that support repayment during a run; their analysis is complementary, microfounding the run margin where we instead discipline a reduced-form hazard with data and push toward an implementable state-contingent rule. We adopt a tractable reduced-form hazard $\mathcal{H}(N/L)$, disciplined by the Jordà et al. (2017) crisis database, that captures this literature's central comparative static (fragility rises as capitalization falls) while remaining embeddable in a globally-solved nonlinear model. Our aim is not a new theory of runs but an analysis of how run risk *interacts* with the real-amplification role of bank capital.

Second, it relates to the literature on *intermediary balance sheets and the financial accelerator*, in which net-worth fluctuations amplify and propagate shocks (Kiyotaki and Moore, 1997; Bernanke et al., 1999; Gertler and Kiyotaki, 2010; Brunnermeier and Sannikov, 2014; He and Krishnamurthy, 2013, 2019), with leverage moving procyclically (Adrian and Shin, 2010; Gertler and Karadi, 2011) and intermediary capital priced as a state variable (Rampini and Viswanathan, 2019). These models stress that strong balance sheets stabilize the financial sector. Our *credit-exposure channel* shares the mechanism but draws the opposite-signed implication for real shocks: the same net worth that buffers fragility raises the scale of intermediation, so better-capitalized banks transmit *larger* productivity shocks. The novelty is that both forces operate through one state variable, bank equity, giving capital a *non-monotonic*, dual role that neither literature isolates on its own.

Third, it contributes to *quantitative analyses of bank capital requirements*. Van den Heuvel (2008) measures the welfare cost of requirements through forgone liquidity provision, and Van den Heuvel and Villalvazo (2026) extend that cost–benefit approach to joint capital and liquidity requirements using sufficient statistics, with regulation rationalized by deposit-insurance moral hazard; in our model the rationale is instead a compositional fragility externality, and the cost of bank distress is macroeconomic (binding leverage constraints and a credit crunch) rather than fiscal. Begenau (2020) shows requirements can raise welfare by reshaping banks' funding and risk choices; Corbae and D'Erasmo (2021) study capital buffers with banking-industry dynamics; and Elenev et al. (2021) embed financially-constrained intermediaries in a general-equilibrium model to evaluate macroprudential policy. These papers share our focus on welfare-relevant capital regulation and the trade-off between resilience and intermediation. Our contribution is to show that the trade-off is non-monotonic once crisis fragility and real-shock exposure operate through the same state variable: requirements that lower crisis risk can simultaneously raise exposure to productivity shocks. We also characterize both the constrained-efficient allocation and an implementable state-contingent leverage rule in a globally solved model that handles the occasionally binding leverage constraint exactly.

Fourth, it speaks to the *macroprudential policy and externalities* literature. A large body of work traces inefficient borrowing to *pecuniary* externalities operating through asset prices and collateral values (Lorenzoni, 2008; Bianchi, 2011; Mendoza, 2010), with optimal macroprudential policy characterized by Bianchi and Mendoza (2018) and the role of nominal rigidities by Farhi and Werning (2016). Our externality is different in kind: it is *compositional*, not pecuniary. Each atomistic bank takes the aggregate failure hazard as given, yet its leverage and equity choices move the sector-wide capital ratio N/L that determines that hazard; no relative price mediates the spillover. The planner corrects it with a Pigouvian wedge on leverage, which we map into a state-contingent leverage rule on bank equity. The closest policy antecedent is the optimized macroprudential debt tax of Bianchi and Mendoza (2018); our instrument instead maps directly to Basel-style capital requirements and is estimated over the nonlinear region around the leverage kink, where the credit-growth triggers of frameworks such as the Basel III countercyclical buffer (Basel Committee on Banking Supervision, 2010) conflate the financial and business cycles.

The remainder of the paper is organized as follows: Section 2 motivates the model with stylized facts, Section 3 presents the model, Section 4 discusses quantitative results, Section 5 analyzes the constrained-efficient allocation and optimal policy rule, and Section 6 concludes.

2 Empirical facts

This section documents three reduced-form facts that motivate the model's two-channel mechanism. Fact 1 disciplines the fragility channel by linking lower system-wide capitalization to a higher probability of banking distress. Facts 2 and 3 discipline the credit-exposure channel: stronger capitalization predicts faster subsequent credit growth, and negative productivity shocks are more contractionary when they arrive after a credit boom. The empirical exercises use annual macro-financial panel data from the historical database of Jordà et al. (2017). For the productivity-shock interaction exercise, we merge the JST panel with Penn World Table TFP growth to identify negative productivity shocks. All three exercises are estimated at the country-year level: Fact 1 with a pooled logit, and Facts 2 and 3 with panel local projections whose baseline design includes country fixed effects, adding year fixed effects, standard macro controls, and stricter inference in robustness designs.

2.1 Bank distress probability and lagged capital ratios

Fact 1: Less-capitalized banking systems have a significantly higher probability of distress. We estimate this relationship with a pooled logit of a pre-crisis indicator on the lagged aggregate capital ratio, using 2,259 country-year observations (154 pre-crisis episodes) from the JST panel. The pre-crisis indicator equals one in the two years immediately preceding a JST systemic banking-crisis onset and zero in tranquil country-years; crisis years themselves are excluded from the estimation sample, following standard early-warning designs. The estimated slope on lagged capital is strongly negative ($\beta = -6.86$ at annual frequency), implying that weaker capitalization is associated with materially higher distress probabilities. Table 1 reports the evidence. At the sample mean capital ratio (15.5%), the fitted pre-crisis probability is 5.8%, and the marginal effect is economically large: a one percentage point increase in the aggregate capital ratio is associated with roughly a 0.38 percentage point reduction in pre-crisis risk. This evidence motivates the hazard block in the model, where fragility decreases with capitalization.

2.2 Healthier balance sheets and subsequent credit growth

Fact 2: Better-capitalized banking systems experience faster subsequent credit growth. We estimate this relationship using country-panel local projections. In the baseline design (country fixed effects with heteroskedasticity-robust inference), the point estimates are positive at every horizon and statistically significant (the cumulative responses carry p-

Table 1: Bank distress risk and lagged aggregate capital ratio: baseline logit summary

n_{obs}	n_{pre}	α (annual)	β (annual)	$\hat{p}$ at mean capital	Marginal effect at mean
2259	154	-1.718 (0.159)	-6.860 (1.207)	0.058	-0.376

Notes: Pooled logit of a pre-crisis indicator on the lagged aggregate bank capital ratio (JST panel). Standard errors in parentheses; both coefficients are significant at the 1% level ($p < 10^{-7}$), with 95% confidence intervals $\alpha \in [-2.03, -1.41]$ and $\beta \in [-9.23, -4.50]$. $\hat{p}$ denotes the fitted pre-crisis probability at the sample-mean capital ratio (15.5%); the marginal effect is $\partial\hat{p}/\partial(N/L)$ at the mean (per unit of the ratio; equivalently about -0.038 per 10 percentage points).

values between 0.001 and 0.039 over horizons 0–5), with the largest responses in cumulative specifications. The estimates attenuate and lose significance under the most demanding design, which adds year fixed effects, lagged controls, and country-clustered standard errors (Online Appendix A), so we read the pattern as robust in sign but sensitive to specification. Figure 1 shows that after an increase in aggregate bank capital ratios, credit-to-GDP rises in subsequent years. This pattern is consistent with the model's credit-exposure channel: stronger bank balance sheets support more credit creation, which can amplify later fluctuations.

$$\Delta_h \text{CreditBoom}_{i,t+h} = \alpha_i^{(h)} + \gamma_t^{(h)} + \beta_h \, \text{CapRatio}_{i,t-1} + \Gamma_h' Z_{i,t-1} + \varepsilon_{i,t+h}, \tag{1}$$

where i indexes countries, t years, and $h \in \{0, \dots, 5\}$ horizons. The dependent variable is credit growth relative to GDP, measured under alternative windows (3-year and 5-year definitions) and reported both in annual and cumulative form. In the figure labels, `dlog_c2g_3y` denotes cumulative 3-year credit-to-GDP growth, while `dlog_c2g_5y` denotes cumulative 5-year credit-to-GDP growth. $\alpha_i^{(h)}$ are country fixed effects and $\gamma_t^{(h)}$ are global year fixed effects; the year effects and the lagged macro controls $Z_{i,t-1}$ enter in the robustness designs (ii)–(iii) below, while the baseline design (i) includes country fixed effects only.

Figure 1 summarizes the result visually. Across specifications, the estimated *cumulative* effect of capital on subsequent credit growth is positive at every horizon: economies with stronger bank capital positions experience larger subsequent credit-to-GDP expansions. (The annual counterparts in the most conservative design turn marginally negative at long horizons; Online Appendix A.) In the baseline design the estimates are statistically significant at all horizons; significance is lost not at longer horizons but once year fixed effects, lagged controls, and country-clustered inference are added (the most conservative design, reported in Online Appendix A). We report three core specification setups: (i) country fixed effects only with HC1 inference (the baseline design), (ii) country plus year fixed effects with lagged controls and HC1 inference, and (iii) the same country-plus-year fixed-effects-with-controls specification with country-clustered standard errors. In designs (ii)–(iii) the identifying variation comes from within-country fluctuations in capitalization after netting out common global shocks.

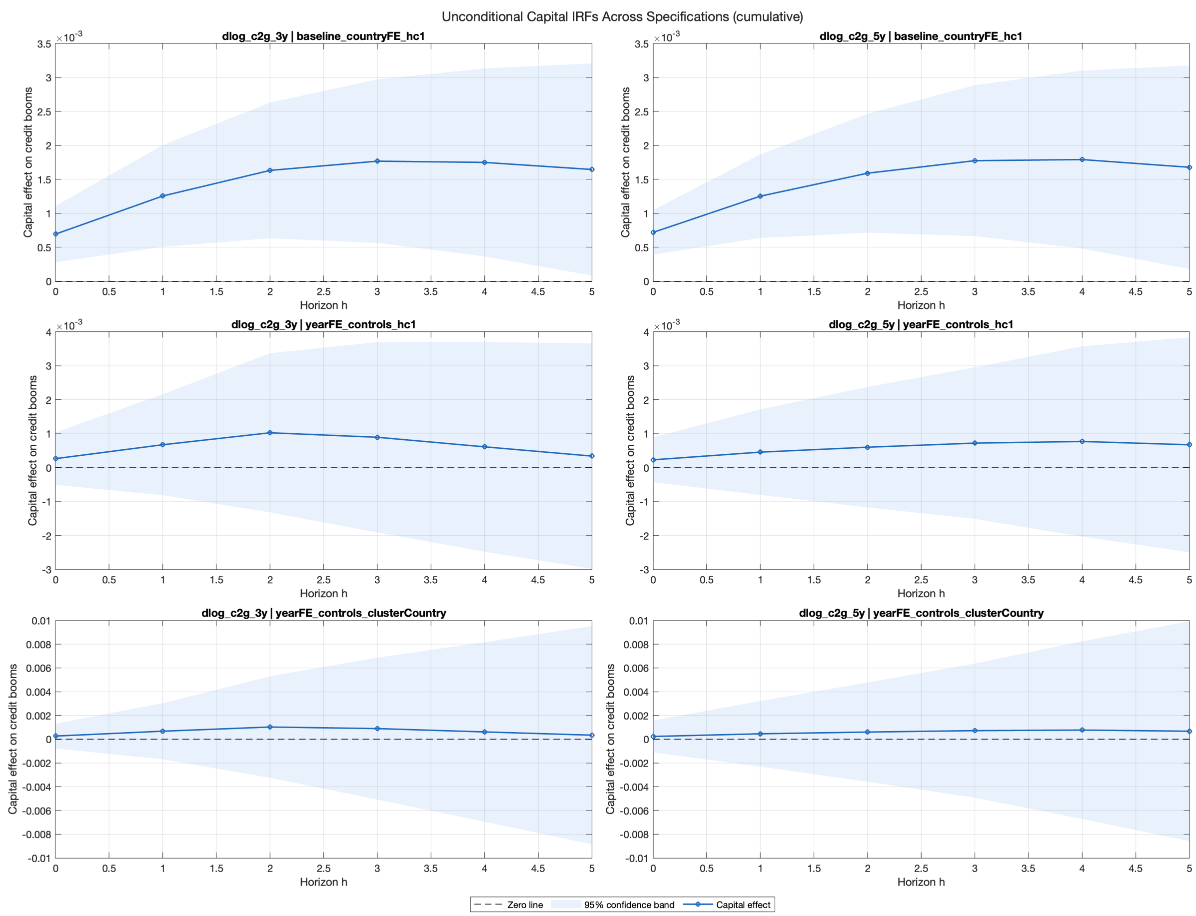

Figure 1: Panel LPs: cumulative response of credit-boom measures to lagged aggregate bank capital ratios

Notes: The figure overlays cumulative local-projection coefficients by horizon across alternative fixed-effects, controls, and inference specifications in the JST panel. The horizontal axis reports the horizon in years after the initial capital-ratio observation; the vertical axis reports the estimated response of credit-to-GDP growth. In the plot labels, `dlog_c2g_3y` denotes cumulative 3-year credit-to-GDP growth and `dlog_c2g_5y` denotes cumulative 5-year credit-to-GDP growth. The annual IRF counterpart is reported in Online Appendix A.

2.3 Credit booms and the output effects of negative productivity shocks

Fact 3: Negative productivity shocks are more contractionary when they hit after a credit boom. In the data, downturns that begin from high-credit-growth states are systematically deeper over subsequent years, indicating that prior balance-sheet expansion amplifies the real effects of adverse shocks. We estimate this pattern using interaction local projections.

$$\Delta_h y_{i,t+h} = \alpha_i^{(h)} + \gamma_t^{(h)} + \theta_h \mathbb{1}\{\Delta a_{i,t} < \tau\} + \psi_h \text{Boom}_{i,t-1} + \delta_h \Big(\mathbb{1}\{\Delta a_{i,t} < \tau\} \times \text{Boom}_{i,t-1} \Big) + \Pi_h' X_{i,t-1} + u_{i,t+h}, \tag{2}$$

where $\Delta_h y_{i,t+h}$ is horizon-h output growth, $\mathbb{1}\{\Delta a_{i,t} < \tau\}$ identifies negative TFP shocks from PWT data, and $\text{Boom}_{i,t-1}$ is a lagged credit-boom indicator from the JST panel. We study two shock thresholds ($\tau = 0$ and lower-quintile TFP growth), annual and cumulative outcomes, and alternative boom definitions.

Figure 2 provides evidence in policy terms: when negative productivity shocks occur after a credit boom, output tends to fall by more over subsequent years than when the same shocks occur outside boom states. This is exactly the amplification pattern implied by the model's exposure channel.

As in Section 2.2, we report three core specification setups: baseline country fixed effects with HC1 inference, country-plus-year fixed effects with controls and HC1 inference, and country-plus-year fixed effects with controls and country-clustered inference. Controls include lagged macro-financial conditions to isolate differential amplification rather than average business-cycle variation. The interaction coefficient δ_h is the object of interest: it measures whether the output cost of a negative productivity shock is larger after a credit boom.

For boom definitions, we use high-credit-growth indicators based on the upper quantile of the pre-shock credit-to-GDP growth distribution, with alternative windows for credit accumulation (3-year and 5-year growth horizons). The baseline definition is the top-quintile 3-year credit-growth state; robustness replaces this with the corresponding top-quintile 5-year growth state.

Figure 2 reports cumulative responses under the $\tau = 0$ shock definition. Across baseline and robustness specifications, δ_h is negative over medium horizons, implying that output losses are deeper when adverse productivity shocks arrive in high-credit-boom states. This is the empirical counterpart of the model's credit-exposure channel.

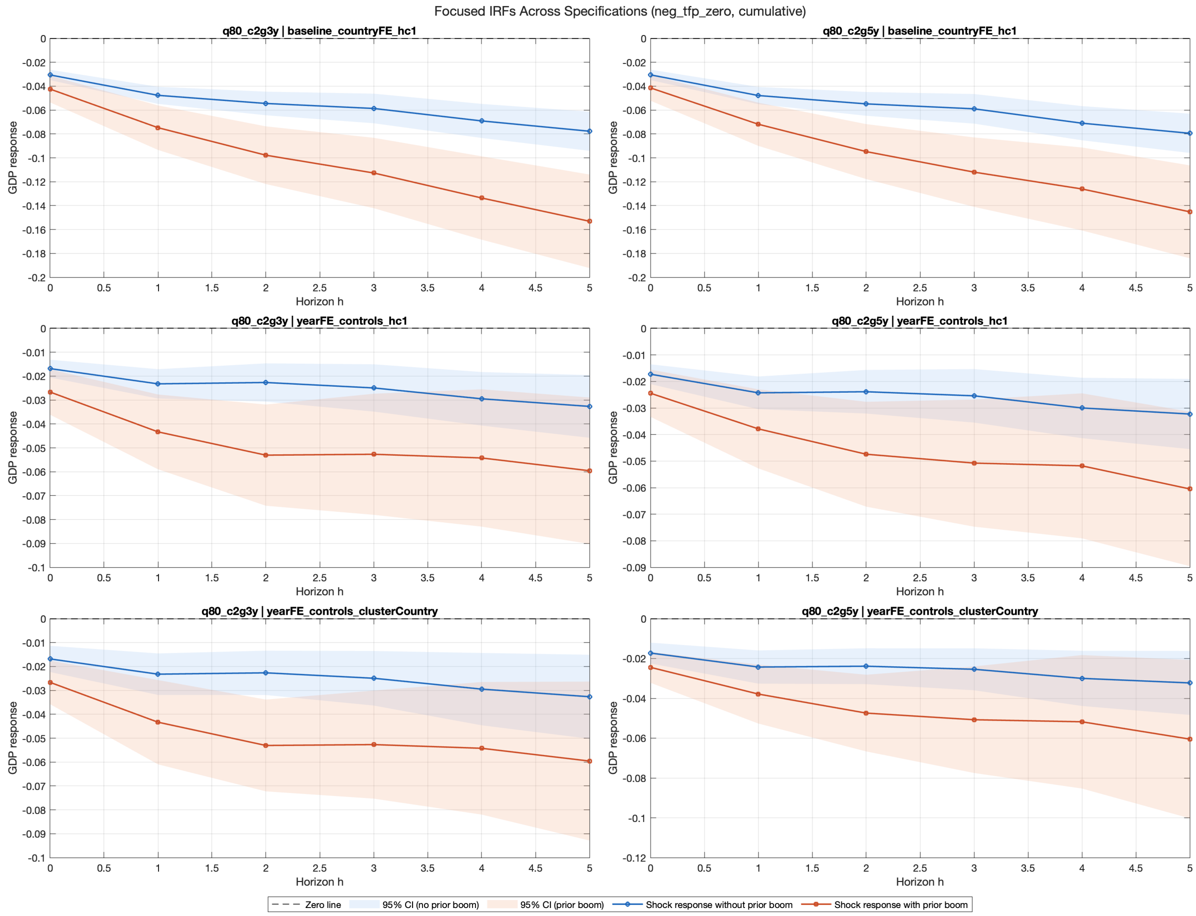

Figure 2: Panel LPs: cumulative output response to negative TFP shocks, conditional on prior credit-boom state

Notes: The figure plots the estimated cumulative output response to a negative TFP shock along two conditional paths (with and without a prior credit boom) across specification variants. The horizontal axis reports the horizon in years after the shock; the vertical axis reports the cumulative output response. The interaction coefficient δ_h in equation (2) is the vertical gap between the with-boom and without-boom paths: the with-boom path lying below means output falls more after adverse productivity shocks when they follow a credit boom. TFP denotes total factor productivity. The three alternative IRF variants are reported in Online Appendix A.

Taken together, these results discipline the model by pinning down both the objects we calibrate and the mechanisms we emphasize. First, Fact 1 maps directly into the fragility hazard in Section 3.3: the estimated logit slope disciplines the slope of $\mathcal{H}(X)$ in Equation 6 directly, and the hazard intercept is then set so that the model's crisis probability at its own steady-state capital ratio matches the empirical pre-crisis probability of roughly 5–6% observed at the sample mean. Second, Facts 2 and 3 discipline the model's credit-exposure mechanism. Fact 2 supports the idea that stronger bank balance sheets permit larger balance-sheet expansion, which in the model operates through the bank's lending choice and the leverage constraint in Equations 4 and 5. Fact 3 then disciplines the quantitative importance of that mechanism by showing that downturns are more severe after credit booms, matching the model's prediction that high lending states amplify negative productivity shocks through the same intermediation channel. In this sense, the empirical section does not just motivate the model qualitatively: it anchors the hazard block and identifies the balance-sheet channel that drives the model's state-dependent amplification results in Sections 4 and 5.

3 Model

Before turning to the formal structure, the model can be understood in simple terms as combining two opposing effects of bank equity. On one side, low equity makes banks fragile: when the banking sector is highly leveraged, the probability of a failure, run, or funding stress rises, so scarce equity amplifies financial crises. This is the fragility channel. On the other side, high equity relaxes banks' balance-sheet constraints and supports more lending in good times; that larger credit exposure raises output when fundamentals are strong but makes the real economy more sensitive when adverse productivity shocks arrive. This is the credit-exposure channel. The capitalization dilemma comes from the fact that the same policy lever - bank capital - moves both forces at once: more equity insures the economy against financial crises, but it can also increase exposure to real-sector downturns by enabling larger balance sheets before the shock.

3.1 Environment

Time is discrete and infinite. The economy consists of a representative household, a unit continuum of atomistic banks, and a representative firm. The aggregate state is $X = [N, A]$, where N is aggregate bank equity and A is aggregate TFP. We use lowercase letters (e.g., n, l, b) for individual bank variables. Each bank is of measure zero and takes the aggregate capital ratio N/L, and hence the failure hazard $\mathcal{H}(X)$, as given; we study the symmetric equilibrium, in which individual choices coincide with their aggregate counterparts ($n = N$, $l = L$, $b = B$). Because each atomistic bank ignores its own contribution to the aggregate N/L, the decentralized equilibrium is inefficient; this is the source of the externality the planner corrects in Section 3.7.

3.2 Bank

The bank maximizes its franchise value:

$$\begin{aligned} J(n;X) = \max_{n'} \; & \pi(n;X) + n - n' \\ & + (1 - \mathcal{H}(X))\, \mathbb{E}\, q(X,X')\, J(n';X') \\ & + \mathcal{H}(X)\, \mathbb{E}\, q(X,X')\, J\big((1-\varphi)n';X'\big) \end{aligned} \tag{3}$$

where $\pi(n;X)$ denotes bank profit, $q(X,X')$ is the stochastic discount factor, $\mathcal{H}(X)$ is the endogenous fragility hazard rate, and φ is the fraction of equity destroyed in a failure. The two continuation values embed the dual nature of the bank's environment: with probability $1-\mathcal{H}$, the bank enters the next period intact; with probability $\mathcal{H}$, it suffers an equity loss that degrades its future intermediation capacity.

Bank profit is:

$$\pi(n;X) = \max_{l} \; \Phi(l)(R^K(X)-1) - (R^B(X)-1)(l-n) \tag{4}$$

where $\Phi(l) = \Phi_A l^{\Phi_B}$ is a concave intermediation technology ($\Phi_B < 1$), R^K is the gross return on capital, R^B is the gross deposit rate, and l is total lending. Deposits are $b = l - n$. The concavity captures diminishing returns in intermediation: each additional unit of funds raised installs less additional productive capital, $\Phi'(l) > 0 > \Phi''(l)$, so the marginal product of lending declines with scale. The capital installed is purchased out of output, as the accounting identity below makes explicit; the only real cost of intermediation itself is the deposit-taking cost $\xi(b)$ introduced below.

The bank faces a leverage constraint:

$$b \leq \phi_0 + \phi_1 n \tag{5}$$

with associated multiplier $\lambda \geq 0$.

Intuitively, this constraint turns equity losses into a nonlinear lending response. When the constraint is slack, a small decline in bank equity can be absorbed mostly through balance-sheet adjustment; once the constraint binds, the same loss forces deposits and lending to contract one-for-one, creating the kink that later appears as the economy's "fault line".

3.3 Fragility hazard

The failure probability is:

$$\mathcal{H}(X) = \sigma\left(\kappa_1 \frac{N}{L(X)} + \kappa_0\right) \tag{6}$$

where $\sigma(\cdot)$ is the logistic sigmoid, N/L is the aggregate capital ratio, and $\kappa_1 < 0$ so that lower capitalization raises fragility. Thus, a lower aggregate capital ratio N/L - i.e., a more highly leveraged banking system - sharply raises the probability of a crisis, consistent with Fact 1. Intuitively, a thinly capitalized banking sector has less capacity to absorb losses and is more vulnerable to depositor withdrawals, asset fire sales, or interbank contagion—all of which are captured in reduced form through the capital ratio. We use a logistic hazard because it is parsimonious, maps naturally into probabilities bounded between zero and one, and can be disciplined directly by the logit evidence in Section 2.1. Alternative monotone hazard forms would preserve the qualitative mechanism as long as fragility rises steeply when capitalization falls; the logistic specification provides the most direct empirical mapping while keeping the global solution tractable. The parameter κ_1 is calibrated from a logit regression of crisis probability on the capital ratio using the Jordà et al. (2017) macrohistory database.

3.4 Firm

A representative firm operates a Cobb-Douglas technology:

$$Y = AK^{\alpha}(H^D)^{1-\alpha} \tag{7}$$

and hires capital and labor competitively. With full depreciation, bank equity is the sole aggregate endogenous state, which keeps the global solution tractable.

3.5 Household

The representative household has GHH preferences with a general CRRA parameter γ:

$$u(C, H^S) = \frac{\left(C - \eta \frac{(H^S)^{1+1/\chi}}{1+1/\chi}\right)^{1-\gamma}}{1-\gamma} \tag{8}$$

with discount factor β, Frisch elasticity χ, and labor disutility weight η. In the baseline calibration, $\gamma = 1$, so the utility function reduces to $u = \log\big(C - \eta(H^S)^{1+1/\chi}/(1+1/\chi)\big)$.

The household budget constraint is:

$$C + p'(X)\, a' = p(X)\, a + \left(R^B(X) - 1\right) B + w(X)\, H^S - \xi(B) \tag{9}$$

where $\xi(B) = \zeta_1 B + \zeta_2 B^2$ is a portfolio adjustment cost for deposits, $p(X) = J(N; X)$ is the (cum-dividend) bank equity price, $p'(X)$ is the ex-dividend price at which new holdings a' are purchased—pinned down by the household's first-order condition as the discounted expected cum-dividend value, $p'(X) = \mathbb{E}\left[q(X, X')\, p(X')\right]$, with the expectation weighting the no-run and run branches by $1 - \mathcal{H}(X)$ and $\mathcal{H}(X)$—and a is equity holdings. Deposits are intra-period: the household earns the within-period spread $(R^B(X) - 1)\, B$ on funds placed with the banking sector, so only $\xi(B)$ is a real resource cost (in equilibrium, with $a = 1$, this is the consumption form $C = (R^B(X) - 1)\, B + d + w(X)\, H^S - \xi(B)$ used in the solution, where d is the bank dividend). When a failure occurs (probability $\mathcal{H}(X)$), a fraction φ of bank equity is destroyed between periods, mapping the chosen retention $n'(N; X)$ into realized equity $(1 - \varphi)\, n'(N; X)$; the lost equity is rebated to no one, so bank failures destroy aggregate wealth, with the real cost accruing through lower future intermediation (the timing is made precise below equation (11)).

3.6 Equilibrium

Definition 1 (Recursive Competitive Equilibrium)**.** *A recursive competitive equilibrium is a collection of: (i) a bank value function $J(n; X)$ and policy functions $\{n'(n; X), l(n; X)\}$; (ii) a household value function $v(a; X)$ and policy functions $\{a'(a; X), B(a; X), H^S(a; X)\}$; (iii) pricing functions $\{R^K(X), R^B(X), w(X), p(X), p'(X), q(X, X')\}$; and (iv) an aggregate law of motion $N' = \Gamma(X, \varepsilon)$; such that: (a) the bank's and household's policy functions solve their respective Bellman equations; (b) all markets clear ($\Phi(L) = K$, $B(1; X) = l - n$, $a'(1; X) = 1$, $H^S = H^D$); (c) the aggregate law of motion is consistent with individual decisions; and (d) prices are consistent with optimality.*

In equilibrium, all markets clear. The key aggregate conditions are:

- *Equity transition.* Aggregate bank equity evolves according to

$$N'(X) = \begin{cases} n'(N; X) & \text{with prob. } 1 - \mathcal{H}(X) \\ (1 - \varphi)\, n'(N; X) & \text{with prob. } \mathcal{H}(X) \end{cases} \tag{10}$$

- *National accounting.* The national accounting identity, stated in terms of the bank's *chosen* retention n', is

$$Y = C + \xi(B) + K + (n' - N) \tag{11}$$

where C is consumption, $\xi(B)$ is a real resource cost (deadweight loss from intermediation), and bank dividends $d = \pi + n - n'$ flow to households as equity income.

The timing of the failure loss deserves precision. Identity (11) holds *every* period: within the period, all output is accounted for by consumption, the intermediation cost, capital, and chosen retention, so no goods are destroyed contemporaneously by a failure. The failure shock realizes *between* periods, mapping the chosen retention n' into realized equity $N' = (1-\varphi)n'$ via the transition (10). The destroyed wedge $\varphi n'$ is therefore a loss of the *wealth stock* carried into the next period (not a transfer, since it is rebated to no one), and its real cost accrues *going forward*: lower equity means less intermediation, capital, and output in subsequent periods (amplified when the loss pushes the bank into the constrained region). In this sense bank failures destroy aggregate resources in present value even though no current-period goods vanish at the moment of failure.

- *Stochastic discount factor.* The SDF is

$$q(X, X') = \beta \frac{u_c(C', H^{S'})}{u_c(C, H^S)} = \beta \frac{\widetilde{C}(X)}{\widetilde{C}(X')} \tag{12}$$

where $\widetilde{C} := C - \eta (H^S)^{1+1/\chi}/(1+1/\chi)$ is composite consumption and the second equality holds under log-GHH preferences ($\gamma = 1$, so $u_c = 1/\widetilde{C}$).

3.7 Social planner's problem

The constrained social planner (SPP) solves the same problem as the bank but internalizes the effect of aggregate equity on the hazard rate $\mathcal{H}(X)$. The bank's first-order condition for n' (the Euler equation) is:

$$1 = (1 - \mathcal{H})\, \mathbb{E}\left[q\, J_1(n'; X')\right] + \mathcal{H}\,(1-\varphi)\, \mathbb{E}\left[q\, J_1\big((1-\varphi)n'; X'\big)\right] \tag{13}$$

where the envelope condition gives $J_1(n; X) = R^B(X) + \lambda(1 + \phi_1)$. The planner internalizes that aggregate leverage raises the failure hazard, which introduces a Pigouvian wedge. On the deposit margin this wedge is

$$\tau(X) = \frac{\beta}{u_c(C, H^S)} \cdot \frac{\partial \mathcal{H}}{\partial B} \cdot \mathbb{E}\left[V(n'; X') - V\big((1-\varphi)n'; X'\big)\right], \tag{14}$$

where $V(N; X) := v(1; [N, A])$ is the household's equilibrium lifetime utility as a function of the aggregate state, and $\partial\mathcal{H}/\partial B > 0$ reflects that higher leverage increases the failure hazard (since $\partial(N/L)/\partial B = -N/L^2 < 0$ and $\kappa_1 < 0$). Because $V(n') > V((1-\varphi)n')$, the wedge $\tau > 0$. The modified lending FOC is therefore $\Phi'(l)(R^K - 1) - (R^B - 1) = \lambda + \tau$, inducing lower leverage.

The marginal *social* value of equity is larger still. A unit of retained equity not only earns the deposit-margin correction but also crowds out deposits one-for-one and so expands the capital ratio N/L against which the hazard is assessed; the planner's equity envelope therefore carries the *scaled* wedge $\nu(X) = \tau(X) \cdot L/N$:

$$J_1^{\text{SPP}}(n; X) = R^B(X) + \lambda(1 + \phi_1) + \nu(X), \qquad \nu(X) = \tau(X)\,\frac{L}{N}. \tag{15}$$

Since $L/N > 1$, the equity wedge exceeds the deposit wedge, $\nu > \tau$: the planner values bank equity by more than a deposit tax alone would imply. This distinction is quantitatively decisive: a scheme that corrects only the deposit margin (a τ-only Pigouvian tax) leaves banks under-capitalized relative to the household optimum, recovering only a fraction of the available welfare gain.[2] Online Appendix C derives ν from the planner's envelope condition. The underlying externality is compositional: each bank takes the aggregate failure probability $\mathcal{H}(X)$ as given, ignoring that its own leverage choice contributes to the sector-wide capital ratio that determines $\mathcal{H}$. The spillover operates through the aggregate quantity N/L rather than a relative price, distinguishing it from the pecuniary externalities of the collateral-constraint literature.

3.8 Analytical properties

The following propositions characterize key properties of the equilibrium. Proofs are provided in Online Appendix C.

Proposition 1 (Sign of the Pigouvian wedge)**.**
Suppose the equilibrium value function $V(\cdot\,; X)$ is strictly increasing in aggregate bank equity, a

[2] We verified this directly: a planner that prices only the deposit margin, adding τ to the lending first-order condition but leaving the equity envelope uncorrected, recovers well under half of the full planner's welfare gain, so the majority flows through the equity wedge ν. The three-channel decomposition of Section 5.6 isolates the contribution of each margin.

property verified in the computed solution. Then, in the constrained social planner's problem, the Pigouvian wedge satisfies $\tau(X) > 0$, and hence the equity wedge satisfies $\nu(X) = \tau(X)\, L/N > 0$, for all X with $\mathcal{H}(X) \in (0,1)$. In the calibrated solution, the planner accumulates more equity than the competitive equilibrium at every state, $n'^{\text{SPP}}(N;X) > n'^{\text{CE}}(N;X)$.

Proof. See Online Appendix C. □

The result follows from two observations: (i) $\partial\mathcal{H}/\partial B > 0$, since $\kappa_1 < 0$ and $\partial(N/L)/\partial B = -N/L^2 < 0$; and (ii) $V(n';X') > V((1-\varphi)n';X')$ for $\varphi > 0$. Together these imply $\tau > 0$, and hence the equity wedge $\nu = \tau\, L/N > 0$. Since $\nu > 0$ enters as $J_1^{\text{SPP}} = R^B + \lambda(1+\phi_1) + \nu > J_1^{\text{CE}}$, the planner's Euler equation requires a higher marginal value of equity than the bank's; in the calibrated solution this is met by retaining more equity at every state.

Proposition 2 (Conditional steady state)**.**
Fix $A \in \{A_L, A_H\}$. If the equilibrium policy function $n'(\cdot\,;A)$ is continuous with $n'(N;A) > N$ for N sufficiently small and $n'(N;A) < N$ for N sufficiently large, then a conditional steady state N_A^{CS} exists. If additionally $n'(\cdot\,;A)$ is nondecreasing and crosses the 45-degree line exactly once—both properties hold in the computed policy functions—the CSS is unique and globally stable under the frozen regime.

Proof. See Online Appendix C. □

The CSS captures the equity level at which dividend payouts exactly offset the bank's retained earnings motive: below N_A^{CS}, equity is scarce and intermediation profits induce accumulation; above it, the marginal return falls below the household's required rate of return and equity is paid out.

Proposition 3 (Planner's CSS ordering)**.**
Under the conditions of Proposition 2, and if the planner retains more equity at every state, $n'^{\text{SPP}}(N;A) > n'^{\text{CE}}(N;A)$ for all N—as holds in the calibrated solution (Proposition 1)—then $N_A^{\text{CS,SPP}} > N_A^{\text{CS,CE}}$ for each TFP regime A.

Proof. See Online Appendix C. □

This follows directly from Proposition 1: since the planner retains more equity at every N, the policy function $n'^{\text{SPP}}(\cdot;A)$ lies above $n'^{\text{CE}}(\cdot;A)$, shifting the fixed point rightward.

4 Quantitative analysis

4.1 Calibration

Table 2 summarizes the baseline calibration. Household preferences follow standard values: discount factor $\beta = 0.96$ (annual), log utility ($\gamma = 1$), unit Frisch elasticity ($\chi = 1$), and labor disutility $\eta = 0.65$, which implies steady-state labor supply of about 0.59 (see the steady-state table in Online Appendix D). The capital share is $\alpha = 0.33$.

For banking parameters, the intermediation curvature $\Phi_B = 0.9$ generates moderate diminishing returns. The quadratic deposit cost $\zeta_2 = 0.5$ pins down the deposit rate spread; the linear component $\zeta_1 = 0$ is set to zero for parsimony. The leverage parameters $\phi_0 = 0.01$ and $\phi_1 = 1.5$ are chosen so that the constraint binds approximately 18% of the time in the CE.[3]

The hazard slope $\kappa_1 = -6.86$ is estimated from the logit regression of the pre-crisis indicator on the lagged capital ratio in Section 2.1, using the Jordà et al. (2017) macrohistory database (the empirical logit intercept is -1.72). Only the slope is taken directly from the data; the model intercept $\kappa_0 = 0.47$ is recalibrated so that the hazard evaluated at the model's steady-state ratio $N/L \approx 0.50$ matches a reference steady-state hazard of about 5%, in line with the empirical pre-crisis probability at the mean capital ratio (5.8%, Section 2.1). The model's *ergodic* crisis frequency is higher (8.9% in the CE, Table 4), since the stationary distribution spends time below the steady-state ratio. The distress loss $\varphi = 0.60$ implies 60% equity destruction upon failure. We set the equity loss in distress to $\varphi = 0.60$ to represent a severe but not total banking sector impairment: the shock destroys a majority of bank equity while allowing the intermediary sector to continue operating, consistent with the model's interpretation of distress as a run, funding stress, or resolution event rather than liquidation.

4.2 Solution method

We solve the model using the Repeated Transition Method (RTM) of Lee (2026b). The key computational challenge is tracing counterfactual paths to evaluate the conditional expectations in the Euler equation (13) accurately. The algorithm proceeds as follows:

[3] The implied steady-state capital ratio $N/L \approx 0.50$ is higher than typical bank capital ratios (5–10%) because, with full depreciation, the bank's balance sheet reflects a single period's lending flow rather than a durable capital stock. The leverage parameters are therefore calibrated to match a binding-frequency target rather than a capital-ratio level; the hazard function is likewise anchored to the model's own steady-state ratio (see below), so the fragility calibration is internally consistent with the model's scale.

Table 2: Baseline Calibration

Parameter	Description	Value
Households		
β	Discount factor	0.960
γ	CRRA coefficient	1.000
η	Labor disutility weight	0.650
χ	Frisch elasticity	1.000
Firms		
α	Capital share	0.330
δ	Depreciation rate	1.000
Banking		
Φ_A	Intermediation scale	1.000
Φ_B	Intermediation curvature	0.900
ζ_1	Deposit cost (linear)	0.000
ζ_2	Deposit cost (quadratic)	0.500
ϕ_0	Leverage intercept	0.010
ϕ_1	Leverage slope	1.500
Fragility		
κ_1	Hazard slope (capital ratio)	-6.86
κ_0	Hazard intercept	0.47
φ	Distress recovery loss	0.60
Aggregate TFP shock		
A	TFP grid	$\{0.98,\ 1.02\}$
Π_A	TFP transition	$\begin{pmatrix} 0.90 & 0.10 \\ 0.10 & 0.90 \end{pmatrix}$

Notes: Lending technology: $\Phi(l) = \Phi_A\, l^{\Phi_B}$. Deposit cost: $\xi(B) = \zeta_1 B + \zeta_2 B^2$. Leverage: $B \leq \phi_0 + \phi_1 N$. Hazard: $\mathcal{H} = \sigma(\kappa_1\, N/L + \kappa_0)$, where $\sigma(\cdot)$ is the logistic sigmoid and $N/L = N/(N+B)$. κ_1 from Jordà et al. (2017) logit (annual, capital ratio).

1. *Backward step.* Given aggregate allocations, the bank's Euler equation (13) is solved for the optimal equity choice $n'(N; X)$ using iso-shock matching, simultaneously determining the household's labor supply and the deposit market equilibrium.

2. *Forward step.* The economy is simulated using the resulting policy functions and aggregate allocations are updated via damped iteration.

3. *Convergence.* The two steps alternate until the mean squared error between successive aggregate allocations falls below 10^{-8}.

The simulation uses $T = 5{,}001$ periods with a burn-in of 500. The deterministic steady state (computed under $\mathcal{H} = 0$) serves as the initial guess; the stochastic equilibrium endogenously determines positive hazard rates.

4.3 Conditional saddle paths

Following Lee (2026a), we analyze the model's equilibrium dynamics through *conditional saddle paths*—the stochastic analogue of deterministic phase diagrams. Fix a TFP regime $A \in \{A_L, A_H\}$ and an initial equity level N_0. The *conditional saddle path* $\mathcal{M}(A; N_0) := \{(N_t, C_t)\}_{t\geq 0}$ is the orbit generated by the policy function $n'(N_t; A)$ and the consumption mapping $C(N_t, A)$ under the frozen regime A—no TFP switches, no bank failures. Crucially, these objects are computed from the decision rules of the *stochastic* equilibrium: agents correctly anticipate future regime switches and bank failures, so the saddle paths embed forward-looking expectations even though the exogenous state is held fixed. The limit point $(N_A^{\text{CS}}, C_A^{\text{CS}})$, the fixed point of $n'(\cdot\,; A)$, is the *conditional steady state* (CSS). Table 3 reports CSS values; Figure 3 plots the saddle paths.

Table 3: Conditional Steady States

		N_{css}	C_{css}
CE (Competitive)	$A = A_L$ (Recession)	0.0305	0.2128
	$A = A_H$ (Expansion)	0.0424	0.2400
SPP (Social Planner)	$A = A_L$ (Recession)	0.0586	0.2140
	$A = A_H$ (Expansion)	0.0729	0.2419

The conditional saddle paths decompose all equilibrium fluctuations into two types of movements. *Across-saddle* jumps occur when the exogenous state switches: since bank equity N is predetermined, a TFP switch from A_H to A_L corresponds to a vertical jump from the expansion saddle to the recession saddle at the same N. *Along-saddle* movements capture the endogenous propagation that follows: once on a given saddle, the economy converges toward the corresponding CSS through the equilibrium policy function. Together, these two forces—exogenous transitions across saddles and endogenous propagation along them—trace out the full stochastic equilibrium path in the (N, C) plane.

The across-saddle gap measures the *credit exposure channel*. The vertical distance

$$\Delta C(N) \;:=\; C(N, A_H) - C(N, A_L) \tag{16}$$

gives the impact effect of a TFP switch on consumption. Because the expansion saddle is

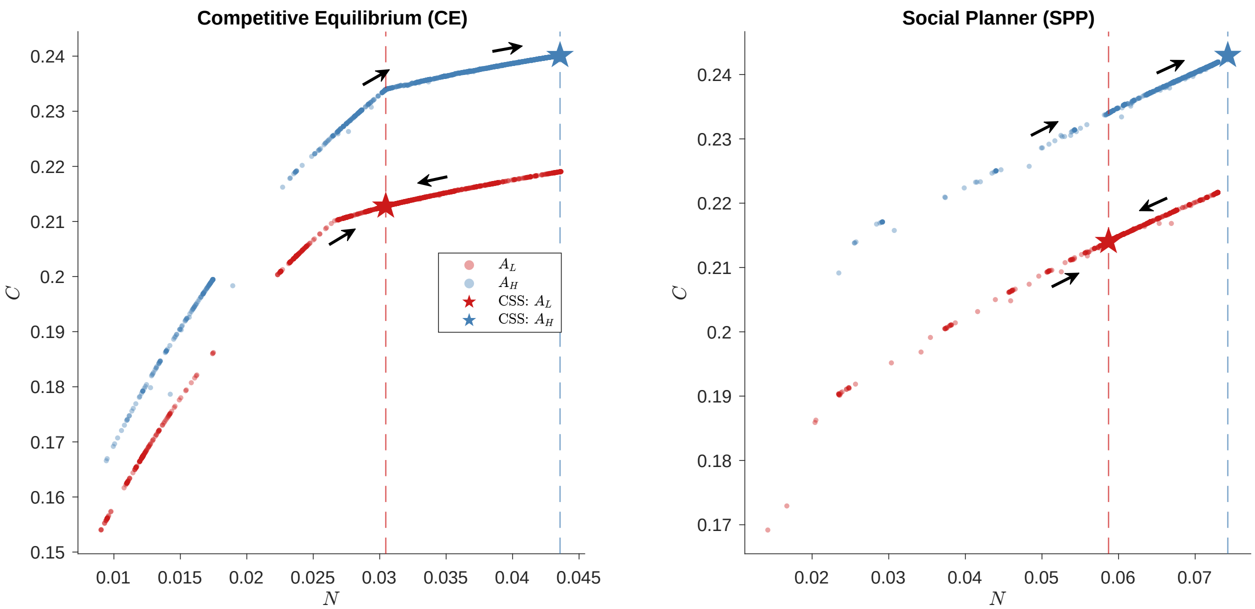

Figure 3: Conditional saddle paths: CE vs. SPP.

Notes: Conditional saddle paths under frozen TFP regimes, computed from the globally solved model. Stars mark conditional steady states. Panel (a): CE. Panel (b): SPP. In the SPP, the planner's recession CSS ($N = 0.0586$) exceeds the CE expansion CSS ($N = 0.0424$).

steeper—higher TFP amplifies the marginal return to bank-intermediated credit—$\Delta C(N)$ widens with N. This generates both size and duration dependence: longer booms push N toward $N^{\text{CS}}_{A_H}$, widening the gap at the point of reversal. Geometrically, the slope difference between the two saddles is the primitive source of state-dependent amplification: if the saddles were parallel, TFP shocks would produce identical consumption drops regardless of capitalization.

The along-saddle dynamics encode the *fragility buffer channel*. A bank failure displaces the economy leftward along the active saddle path by destroying a fraction φ of equity. How much this displacement costs depends on the saddle's local curvature. The occasionally binding leverage constraint (5) creates a kink at intermediate equity levels, dividing the state space into two regimes. Above the kink, equity is abundant relative to the constraint, deposits adjust freely, and losses are absorbed smoothly—the economy operates as if unconstrained. Below the kink, the constraint binds, deposits are rationed, and each unit of lost equity forces a contraction in lending that propagates to output: a credit crunch. The kink is the economy's "fault line"—crossing it transforms a manageable equity loss into a self-reinforcing deleveraging spiral. In the CE, the constraint binds roughly 18% of the time. The planner shifts both saddle paths rightward (Table 3), keeping the economy above the kink in almost every period (binding only 0.2% of the time) and effectively neutralizing this amplification channel.

4.4 Generalized impulse response functions

We compute state-dependent generalized impulse response functions (GIRFs; Koop et al., 1996):

$$\text{GIRF}_t = \mathbb{E}[\text{path}_t \mid \text{shock},\ N_0] - \mathbb{E}[\text{path}_t \mid \text{no shock},\ N_0] \tag{17}$$

expressed as the percentage difference relative to the counterfactual (no-shock) path, with expectations over 1,000 Monte Carlo paths. We condition on N_0 at the 10th percentile (fragile) and 90th percentile (well-capitalized) of the ergodic distribution.

4.4.1 TFP shock

The key message from the TFP experiment is that strong bank balance sheets are a double-edged sword. High equity lowers fragility, but it also supports a larger intermediation volume, so a negative productivity shock hits a more exposed banking system and produces a larger output decline. Figure 4 confirms that well-capitalized banks amplify TFP shocks: starting from high N, the proportional output decline at impact is nearly twice as large as from low N (-10.5% vs. -5.8%). The mechanism is a credit multiplier operating in reverse—the very lending that boosts output in good times magnifies the contraction in bad times. The GIRF compares paths starting in recession (A_L) against expansion (A_H), expressed as the percentage difference relative to the counterfactual (no-shock) path: $100 \times (\text{shocked} - \text{unshocked})/\text{unshocked}$.[4]

4.4.2 Bank hazard shock

The key message from the hazard-shock experiment is the opposite of the TFP case: low capitalization dramatically worsens crisis outcomes. The shock mechanically destroys the same fraction of bank equity in every conditioning state, but its macroeconomic cost depends on whether the economy starts safely above the leverage kink or close to the fault line. Figure 5 shows fragility amplification during expansion: starting from low N, the proportional output decline (-28%) is substantially deeper than from high N (-20%), and the financial side of the collapse is starker still—deposits contract by 48% from low N against 18% from high N, because the equity loss pushes the already-fragile economy below the leverage kink and triggers a credit crunch. Starting from high N, the same

[4] The two paths draw independently from the Markov transition matrix for future TFP but share uniform draws for run realizations to reduce Monte Carlo variance. The counterfactual normalization ensures that the GIRF captures the *proportional* impact of the shock, removing level differences across conditioning states.

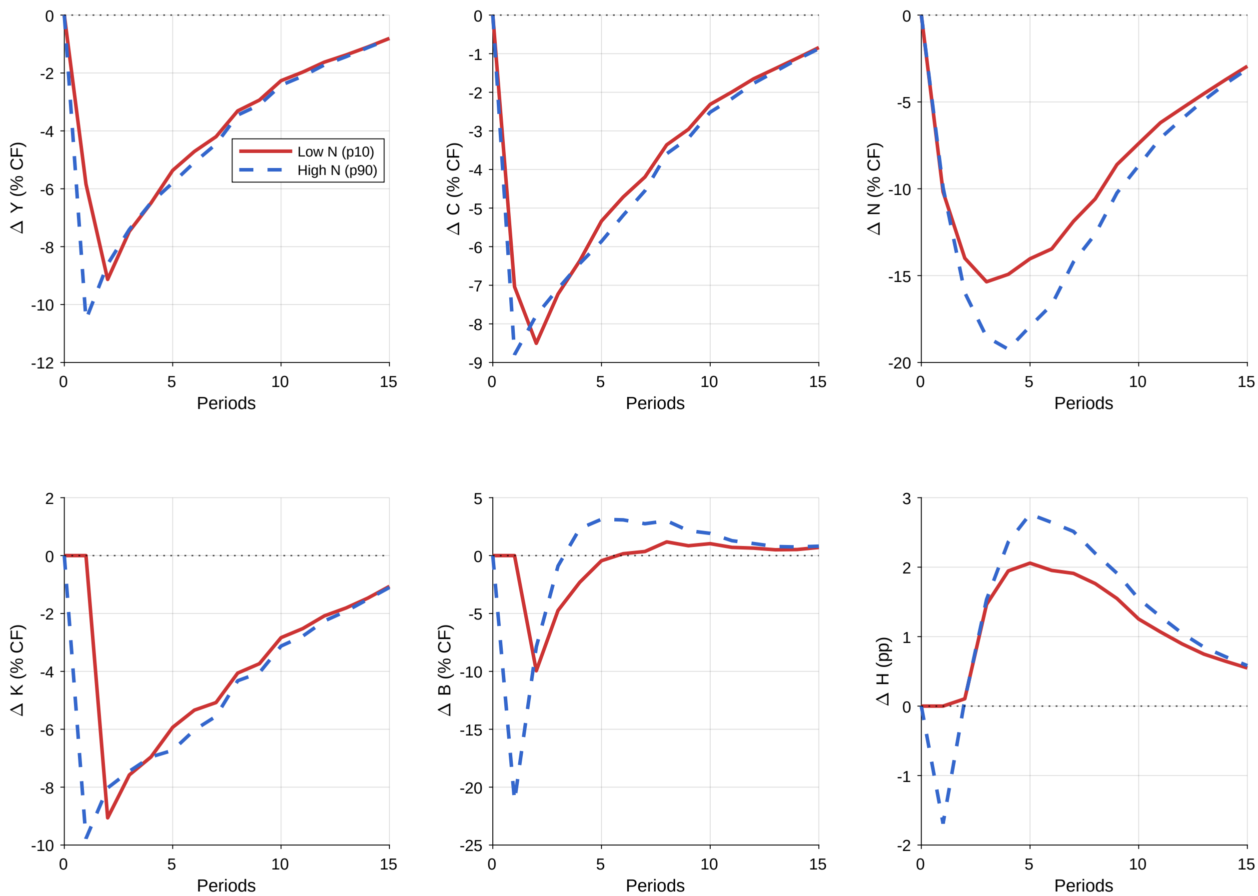

Figure 4: State-dependent GIRFs: negative TFP shock.

Notes: GIRF expressed as percentage difference relative to the counterfactual (no-shock) path. Red solid: low N_0 (p10); blue dashed: high N_0 (p90).

proportional equity loss is absorbed largely within the unconstrained region: the deposit base swings *above* its no-shock path within two periods (+12%) as the well-capitalized bank re-intermediates, though the output recovery is somewhat more drawn out because a larger balance sheet must be rebuilt. Figure 6 shows a similar pattern in recession, with a marginally smaller gap between conditioning states, as banks are already partially deleveraged regardless of initial capitalization. The GIRF isolates this financial channel by forcing a bank failure at $t = 1$ ($\varphi = 60\%$ equity loss), while both paths share the same TFP sequence; from $t \geq 2$, failures are drawn endogenously with shared uniform draws. Under the counterfactual normalization, the equity panel shows a uniform -60% initial drop across all conditioning states, so the figure isolates *recovery dynamics* rather than the mechanical shock itself.

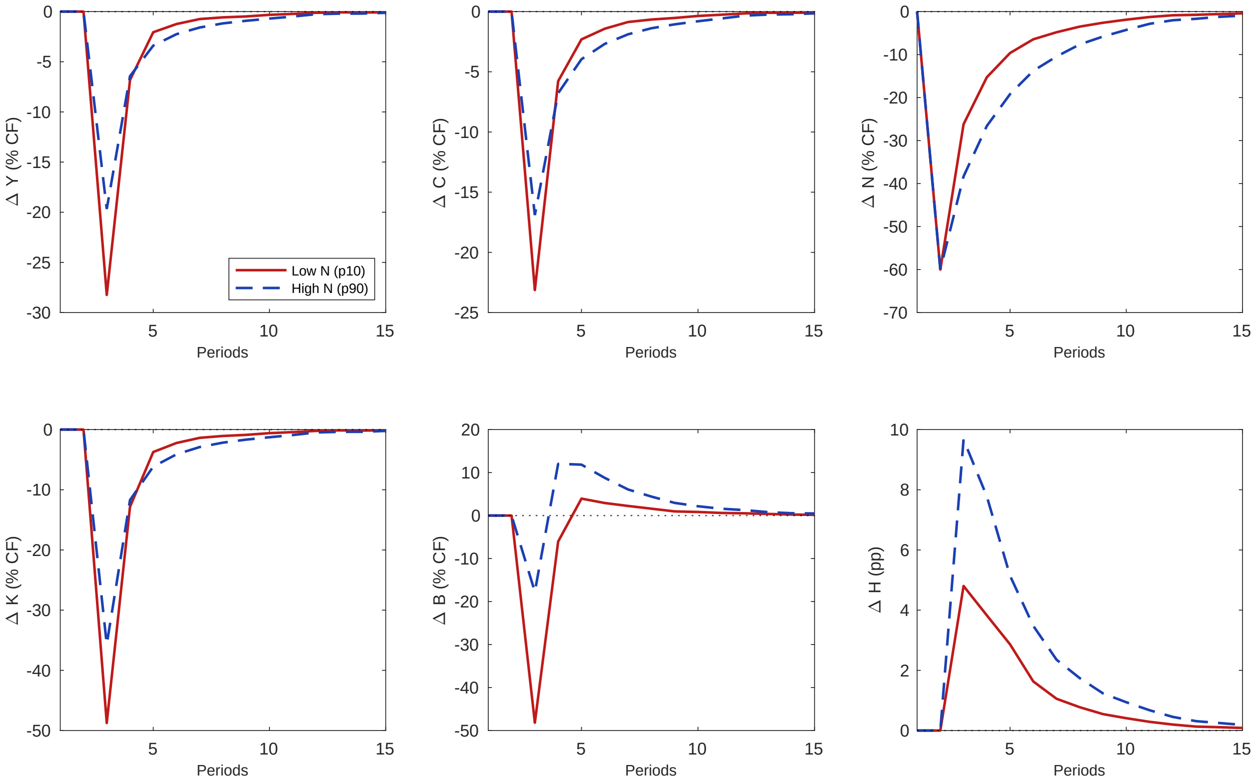

Figure 5: State-dependent GIRFs: bank hazard shock (expansion).

Notes: Both paths share the same TFP sequence starting in expansion. The shock forces a bank failure at $t = 1$. Red solid: low N_0 (p10); blue dashed: high N_0 (p90). GIRF expressed as percentage difference relative to the counterfactual (no-shock) path.

4.4.3 CE vs. SPP

We now overlay the CE and SPP impulse responses to isolate the planner's fragility correction. The plain-language trade-off is this: the planner's crisis-prevention policy makes banks hold more capital in every state, but healthier banks can swing more. As a result, a negative TFP shock causes a slightly larger contraction under the planner when starting from low N - about 9% rather than 6% - which is the cost of the extra safety gained against crises.

Figure 7 compares TFP shock responses, conditioning both CE and SPP on the *same* initial equity levels (the CE's 10th and 90th percentiles) to ensure a like-for-like comparison.[5] Two features emerge. First, in output and consumption (Y, C panels), the CE and SPP responses are similar but not identical: at low N, the SPP's proportional output drop ($\approx -9\%$) is slightly larger than the CE's ($\approx -6\%$). This is the capitalization dilemma in its

[5]If each economy is conditioned on its own percentiles, level differences between CE and SPP confound the comparison. Conditioning on common initial N isolates the effect of the planner's policy function at the same starting point.

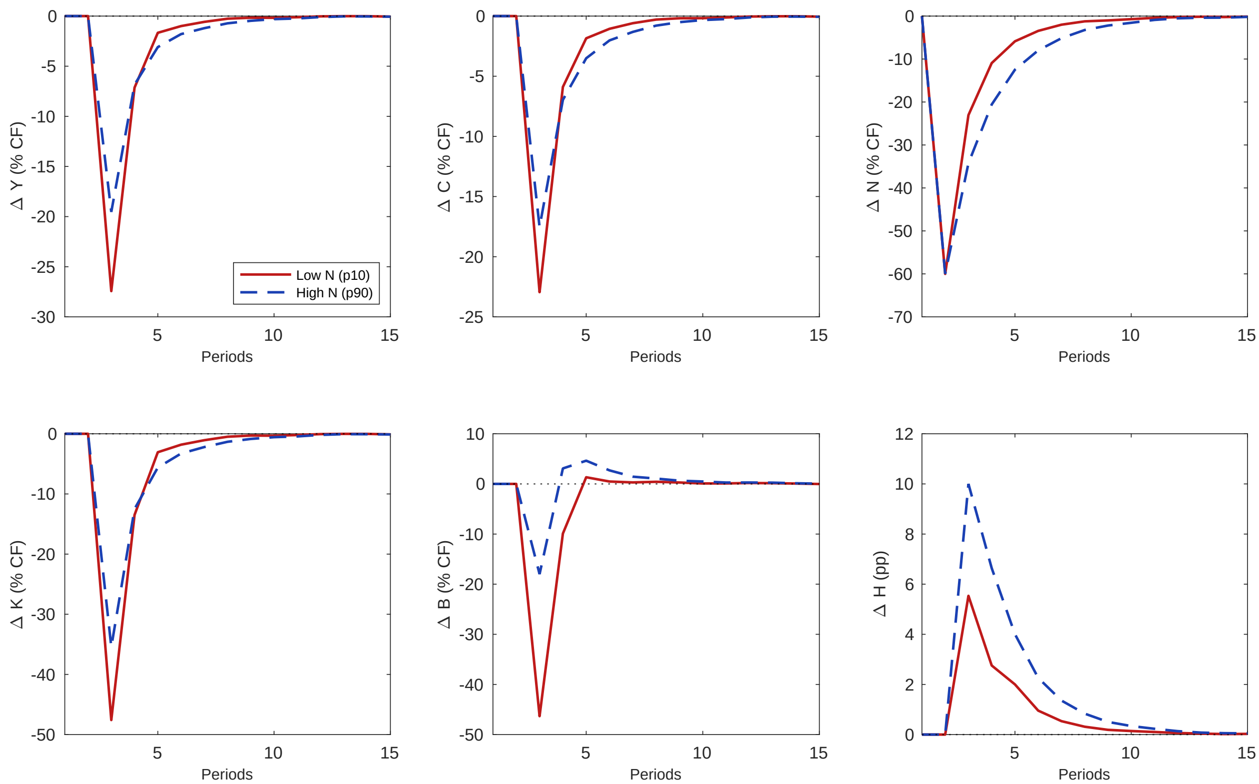

Figure 6: State-dependent GIRFs: bank hazard shock (recession).

Notes: Same construction as Figure 5, with both paths starting in recession.

precise form: at the same equity level, the planner's allocation creates *more* proportional TFP exposure. The mechanism is not that the SPP intermediates more—it actually intermediates slightly less (lower B, lower L)—but that the planner's Pigouvian tax replaces the binding leverage constraint with a smooth price, giving the bank more freedom to adjust its balance sheet in response to TFP. In the CE, the binding constraint acts as a straitjacket that limits both upside and downside; the SPP's voluntary restraint unravels during stress, producing wider swings.

Second, the deposit panel (ΔB) reveals a sharp CE–SPP asymmetry. At both conditioning states, the SPP's deposits respond more dramatically than the CE's—rising in relative terms as the planner's unconstrained bank adjusts freely, while the CE bank's deposits are pinned by the binding constraint. This flexibility is the mirror image of the fragility benefit: the same freedom that allows the planner to stabilize fragility makes the economy more responsive to TFP.

The contrast is stark for financial shocks. Figure 8 overlays the expansion-state hazard shock responses. Under the counterfactual normalization, the equity panel shows a uniform -60% initial drop for all four lines—the mechanical destruction is identical—so the

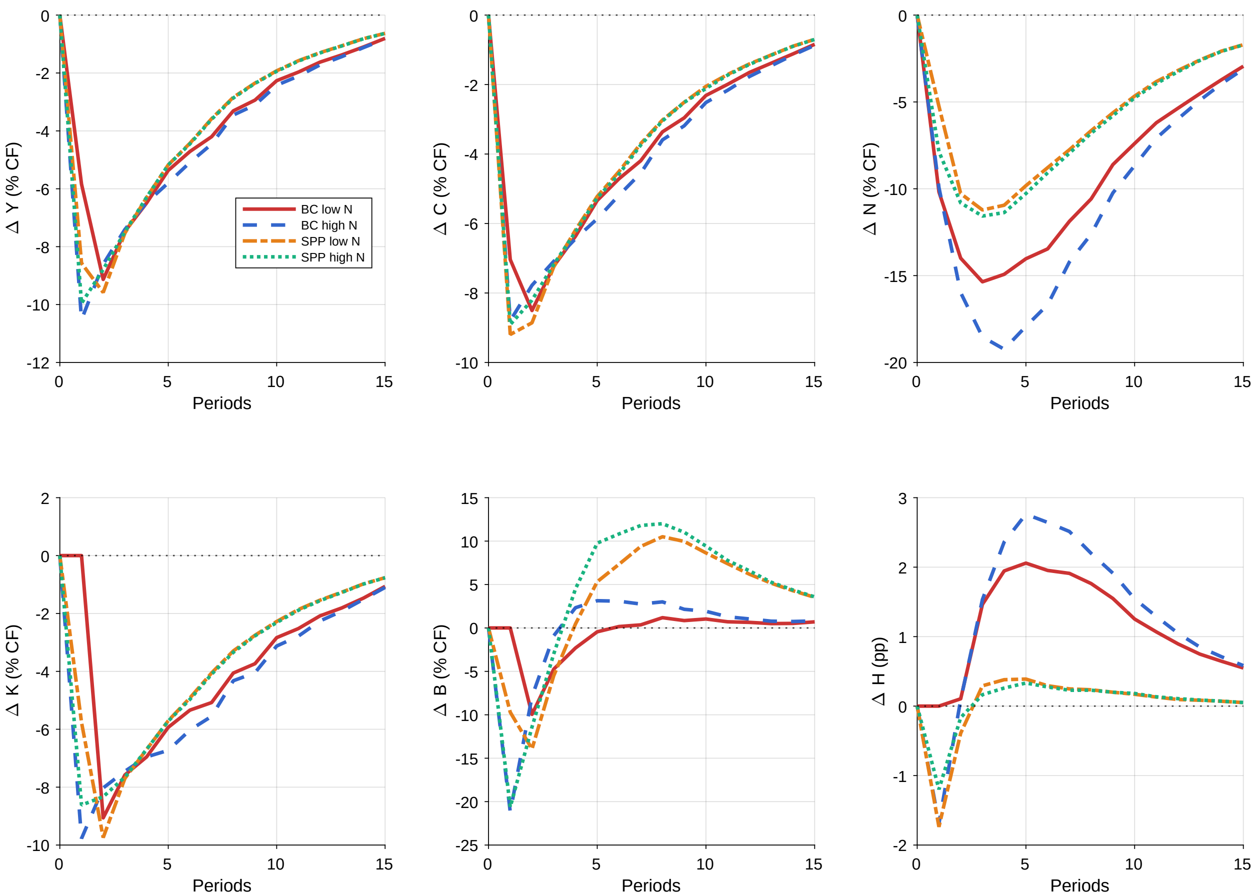

Figure 7: TFP shock GIRFs: CE vs. SPP.

Notes: CE low N_0 (red solid), CE high N_0 (blue dashed), SPP low N_0 (orange dash-dot), SPP high N_0 (teal dotted); the competitive equilibrium is labelled "BC" in the figure legend. Both CE and SPP are conditioned on the CE's 10th and 90th percentiles of N. GIRF expressed as percentage difference relative to the counterfactual (no-shock) path.

figure isolates the recovery dynamics. In the CE, the fragile bank (low N_0) suffers a deep proportional output contraction (-28%) as the leverage constraint binds and deleveraging amplifies the initial equity destruction. The planner's economy, starting from the same conditioning state, absorbs the same proportional equity loss at a fraction of the output cost (-7%): the constraint remains slack (because the Pigouvian wedge has already induced lower leverage), so the bank can rebuild its balance sheet without hitting the hard cap. The deposit panel is particularly striking: CE deposits collapse (-48%), while the planner's deposits, freed from the binding constraint, *expand* rather than contract (peaking around $+27\%$ above the no-shock path) as the bank rebuilds intermediation to exploit the post-crisis recovery.

Figure 9 shows that this pattern persists in recession, though the CE–SPP gap narrows. In recession, lower TFP compresses intermediation profits and both economies

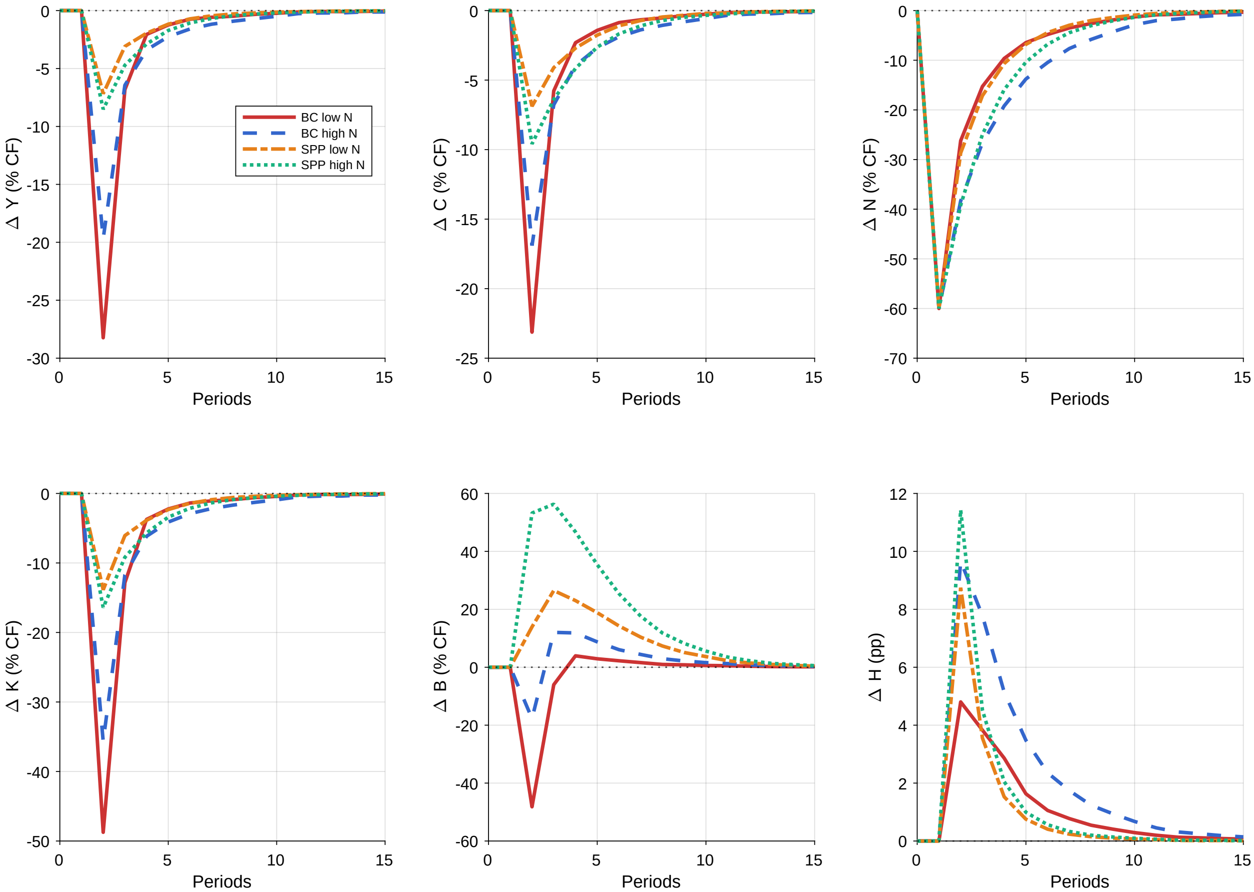

Figure 8: Hazard shock GIRFs: CE vs. SPP (expansion).

Notes: CE low N_0 (red solid), CE high N_0 (blue dashed), SPP low N_0 (orange dash-dot), SPP high N_0 (teal dotted); the competitive equilibrium is labelled "BC" in the figure legend. GIRF expressed as percentage difference relative to the counterfactual (no-shock) path.

have drawn down equity, so even the CE bank is closer to the constrained region—the planner's insurance margin shrinks. Taken together, the three comparisons reveal the capitalization dilemma in its full form. The planner's Pigouvian correction replaces the rigid leverage constraint with a flexible price, which is decisive for financial shocks (the unconstrained bank recovers faster and avoids the credit crunch) but creates wider proportional swings in response to TFP shocks (the unconstrained bank adjusts more freely in both directions). This is the prices-versus-quantities trade-off of Weitzman (1974) applied to macroprudential policy: the price instrument is welfare-superior for the fragility margin but sacrifices the incidental TFP stabilization that the binding quantity constraint provided. Section 5 discusses the mechanisms further.

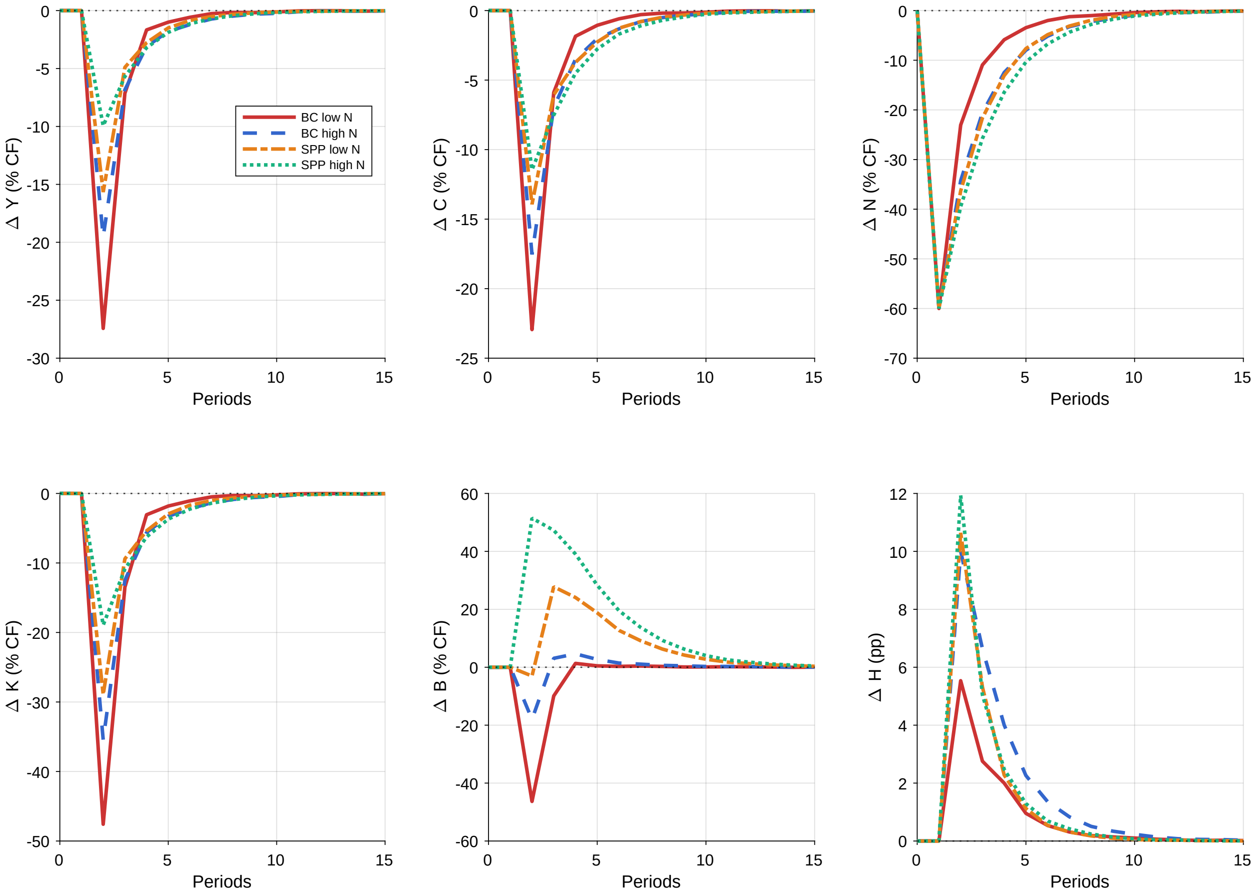

Figure 9: Hazard shock GIRFs: CE vs. SPP (recession).

Notes: Same construction as Figure 8; both paths start in recession.

4.5 Ergodic distributions and welfare

The ergodic distributions are sharply separated: the CE lives in $N \in [0.009, 0.044]$, while the planner spends 99% of its time in $N \in [0.025, 0.073]$, dipping below only in the brief aftermath of its rare runs. The two economies effectively inhabit different, essentially disjoint, regions of the state space, with the planner's everyday equity sitting at or above the CE's best-capitalized states. The planner's recession equity ($N \approx 0.059$ at its conditional steady state) exceeds the CE's *expansion* equity ($N \approx 0.044$), keeping the economy well above the leverage kink even in downturns. The welfare gain is therefore not a pure insurance transfer: by holding more equity the planner suffers far fewer of the destructive runs that periodically wipe out CE capital, so it intermediates more on average and delivers both *higher* mean output and a thinner left tail: higher average consumption *and* lower downside risk at once.

Table 4 reports business cycle moments. The planner maintains much higher mean equity (N/L of 0.78 versus 0.42), lower mean hazard ($\bar{\mathcal{H}}$), and far lower crisis frequency (0.9% vs. 8.9%), and, unlike a blunt cap, it raises mean output and consumption rather than depressing them ($\bar{Y} = 0.335$ vs. 0.326; $\bar{C} = 0.228$ vs. 0.222). A telling difference is in the higher moments: the CE displays strongly negative skewness in Y, C, K, and H (reflecting occasional sharp contractions when the leverage constraint binds), whereas the planner substantially compresses these left tails. Equity N itself is the one exception: its skewness is *more* negative under the planner, because the planner's rare runs cut equity proportionally hard from a high base, but those runs are an order of magnitude rarer. By retaining earnings to build a large equity buffer, the planner all but eliminates the destructive runs that intermittently destroy CE capital; the payoff is a consumption stream that is simultaneously higher on average and less exposed to downside risk. The consumption-equivalent variation (CEV) is the permanent proportional increase in composite consumption $\widetilde{C}_t = C_t - \eta(H_t^S)^{1+1/\chi}/(1+1/\chi)$ that makes the household indifferent between two allocations. We measure it *inclusive of the transition*: both economies start from the same initial condition (equity N_0 equal to the mean of the competitive equilibrium's ergodic distribution, with initial productivity drawn from its stationary distribution), and we compare discounted lifetime welfare $V^P(N_0) = \mathbb{E}_0 \sum_{t\geq 0} \beta^t \log \widetilde{C}_t^P$ under policy P, the expectation averaging over the productivity process and the initial state,

$$\lambda_{\text{CEV}} = \exp\Big[(1-\beta)\Big(V^{\text{SPP}}(N_0) - V^{\text{CE}}(N_0)\Big)\Big] - 1. \tag{18}$$

This is the object the planner actually maximizes: it charges each policy for the transition (the years of retained earnings needed to build the equity buffer from the current position) rather than comparing long-run distributions as if the buffer were already in place. In our calibration $\lambda_{\text{CEV}} = +1.34\%$: moving from the competitive equilibrium to the planner's allocation is worth a permanent 1.34% of consumption.[6]

[6]Because the measure is the planner's own objective, the planner attains the highest value achievable under the leverage technology: any constant cap that improves on the competitive equilibrium is at least as tight as the baseline technology, so its allocation is available to the planner, which therefore weakly dominates it. Section 5.4 shows that no constant leverage cap matches the planner once the transition is priced.

Table 4: Business Cycle Moments: CE vs. SPP

	Competitive Equilibrium			Social Planner		
Variable	Mean	Std. Dev.	Skew.	Mean	Std. Dev.	Skew.
Y	0.3255	0.0311	-1.362	0.3352	0.0201	-0.109
C	0.2216	0.0190	-1.205	0.2279	0.0130	-0.172
K	0.1012	0.0146	-2.124	0.1069	0.0066	-0.301
H	0.5785	0.0288	-1.532	0.5876	0.0176	-0.122
N	0.0334	0.0085	-0.909	0.0651	0.0076	-1.536
B	0.0452	0.0059	-1.492	0.0183	0.0044	$+2.074$
N/L	0.4184	0.0611	-0.131	0.7793	0.0619	-3.046
R^K	1.0743	0.1008	$+3.103$	1.0354	0.0124	$+7.195$
R^B	1.0452	0.0059	-1.492	1.0183	0.0044	$+2.074$
$\mathcal{H}$	0.0888	0.0344	$+0.772$	0.0088	0.0107	$+8.992$
Crisis frequency (%)		8.88			0.88	
CEV (%)		—			$+1.34$	

Notes: Statistics computed over the ergodic distribution ($T = 5001$, burn-in $= 500$). CEV: the transition-inclusive consumption-equivalent gain of moving from CE to SPP (eq. (18)), which starts both economies from the mean of the CE's ergodic equity distribution and prices the transition.

5 Discussion

5.1 The capitalization dilemma

The saddle path decomposition in Section 4.3 maps the two individually well-known channels into a single geometric framework, making their interaction transparent.

The *credit exposure channel*, well documented in the financial accelerator literature, operates through the across-saddle gap $\Delta C(N)$. Higher bank equity supports more intermediation, making the economy more productive but also more exposed to TFP fluctuations. This is a "sowing the seeds" mechanism: credit booms during expansions simultaneously raise output and build vulnerability to productivity reversals. Since $\Delta C(N)$ widens with N, longer booms generate mechanically deeper busts. This channel operates in both the CE and SPP, since both face the same intermediation technology; its strength nonetheless differs across the two, because the planner's slack constraint leaves the bank freer to adjust its balance sheet in response to TFP (the across-saddle gap is somewhat wider for the SPP at the same equity level, as discussed below).

The *fragility buffer channel*, studied in the banking crisis literature, operates through along-saddle displacement and the leverage kink. Lower initial equity means a distress event pushes the economy into the steep, constrained region of the saddle path, where deleveraging amplifies the initial equity loss. This channel is asymmetric across regimes because it depends on the *distance* from the kink, which the planner, but not the decentralized bank, can control. The market failure is a compositional externality: each bank's leverage choice is privately optimal, but the resulting aggregate capital ratio determines the system-wide failure probability that all banks face.

The unified framework reveals a tension that is invisible when the channels are studied separately: bank equity is the single state variable through which both channels operate, and any policy that adjusts it necessarily affects both. This has a direct implication for capital regulation. Tightening leverage requirements (raising ϕ_1) reduces credit exposure by compressing balance sheets and lowering $\Delta C(N)$, but also limits equity accumulation during expansions, potentially leaving the economy more exposed to distress. The regulator faces a frontier, not a free lunch: moving the economy rightward along the saddle path (more equity, less fragility) necessarily steepens the across-saddle gap (more TFP exposure).

In policy terms, the outcome is non-monotonic: extra capital makes rare crisis losses much less likely and less severe, but it can also make the economy more cyclically sensitive to ordinary real shocks, so that the marginal benefit of additional capital eventually comes with a visible stabilization cost.

The nature of this trade-off, however, depends critically on the *instrument* used to correct the fragility externality. The SPP internalizes it on the *lending* margin through a state-contingent Pigouvian wedge $\tau(X)$ that raises the marginal cost of deposits, and on the *equity* margin through the larger wedge $\nu = \tau L/N$ (Section 3.7). On the deposit side this wedge *substitutes* for the leverage constraint: at the planner's allocation, the constraint shadow value λ collapses (from an ergodic mean of 0.043 in the CE to essentially zero in the planner's allocation, which binds only 0.2% of the time), because the wedge has already induced the bank to reduce leverage voluntarily. The bank no longer needs the hard cap to restrain its deposit-taking—the price signal does the work.

This substitution creates a subtle but genuine version of the capitalization dilemma. In the CE, the binding leverage constraint acts as a *straitjacket*: it prevents the bank from expanding in good times, but it also limits how far the bank can contract in bad times. The constraint inadvertently stabilizes the economy's response to TFP fluctuations by compressing the range of intermediation volumes. The planner's Pigouvian tax removes this rigidity. With the constraint nearly slack, the bank is free to adjust its balance sheet in response to TFP—expanding more aggressively in expansions, but also contracting more sharply in recessions. The result is a wider across-saddle gap for the SPP than for the

CE at the same equity level: the proportional output decline from a negative TFP shock is larger under the planner's allocation, not because the planner intermediates more at a given N, but because the bank's *flexibility to respond* is greater when the hard constraint is replaced by a smooth price.

This mechanism connects to a classic insight in the regulation literature. Weitzman (1974) showed that price and quantity instruments yield different welfare outcomes when the relevant functions are nonlinear. Here, the nonlinearity is in the hazard function (6): the marginal cost of leverage rises steeply as the capital ratio falls. The Pigouvian tax, a price instrument, tracks this nonlinearity precisely, providing the optimal correction at every state. The leverage constraint, a quantity instrument, provides a cruder but more rigid correction that happens to stabilize TFP responses as a by-product. The capitalization dilemma is the cost of moving from the rigid instrument to the precise one: the planner gains fragility protection but loses the incidental TFP stabilization that the binding constraint provided.

The dilemma has an important temporal dimension. Ex ante, the planner's allocation is welfare-maximizing: no alternative policy can deliver higher expected lifetime utility. But ex post, conditional on the realized sequence of shocks, the competitive equilibrium can outperform the planner along particular sample paths. During prolonged expansions without bank failures, the CE economy produces more output and higher consumption because it is not holding the extra equity buffer that the planner demands. The planner's higher capital requirement acts as an insurance premium: it depresses dividends and credit intermediation in normal times, and its payoff materializes only when crises actually occur. In a fortunate history where no distress events strike, the premium is paid but never collected. This ex-post reversal is not a deficiency of the planner's policy but a defining feature of insurance: any policy that reduces tail risk must sacrifice expected payoffs in the states where the insured event does not occur. For the central banker, this creates a political economy challenge: the costs of higher capital requirements are visible in every period, while the benefits are concentrated in rare crises that may not materialize during a policymaker's tenure.

5.2 Precautionary savings as macroprudential policy

The planner's equity Euler differs from the bank's by the equity wedge $\nu(X) = \tau(X)\, L/N$, built on the Pigouvian object (14), which prices the marginal social value of equity as a fragility buffer. This wedge generates three reinforcing consequences. First, the planner's conditional equity targets are far higher: its recession conditional steady state ($N \approx$

0.059) nearly doubles the CE's (0.030), and in expansions the planner keeps accumulating to a still higher ergodic level ($N \approx 0.069$ versus 0.044). Second, these higher equity buffers prevent distress events from triggering the leverage constraint; deposits do not contract; the ΔB GIRF turns *positive* as the unconstrained bank rebuilds intermediation (Section 4.4.2), averting the credit crunch that drives CE output losses. Third, even immediately after a run the planner's equity stays near the leverage kink ($N \approx 0.026$) rather than collapsing through it as in the CE ($N \approx 0.009$), so the economy recovers along the flat portion of the saddle path without the amplification that prolongs CE recessions.

These mechanisms are mutually reinforcing: higher equity targets enable deposit stabilization, which prevents the constraint from binding and ensures fast recovery. In the CE, the externality operates through a tragedy-of-the-commons logic: each bank rationally minimizes its own equity to maximize dividends, but the collective outcome is a fragile system in which every bank is exposed to crises that none would individually choose.

The calibrated model pins down the implied Pigouvian wedge. The deposit-margin object $\tau(X)$ enters in the same units as a net interest rate: it is the per-unit tax on deposits that prices the fragility externality on the bank's *lending* margin. (The planner's *equity* margin carries the larger wedge $\nu = \tau L/N$ of Section 3.7; we return to what a deposit tax alone can and cannot achieve below.)

Table 5: Implied Pigouvian Tax

	τ
Mean	0.0225
$A = A_L$ (Recession)	0.0229
$A = A_H$ (Expansion)	0.0221
Std. Dev.	0.0090

Notes: The Pigouvian wedge $\tau(X)$ from (14), in net-rate units (same as $R^B - 1$). Statistics over the ergodic distribution.

The wedge is countercyclical, rising in recessions, when the capital ratio is low and the hazard function is steep, and falling in expansions. The key insight is what this tax *buys* in terms of the saddle-path geometry. The CE's recession CSS ($N = 0.0305$) sits right at the leverage kink, so a single distress event pushes the economy into the steep, constrained region where deleveraging spirals take hold. The planner's wedge shifts its equity targets well above this zone: the planner's recession CSS ($N = 0.0586$) exceeds even the CE *expansion* CSS ($N = 0.0424$). The planner's worst-case anchor point already lies beyond the best-case anchor of the unregulated economy.

This separation from the kink is not a knife-edge property of the calibration but arises from a self-reinforcing mechanism in the structure of τ itself. The Pigouvian wedge is *endogenously amplifying*: it rises sharply as the economy approaches the constrained region, because proximity to the kink simultaneously steepens the hazard function $\partial\mathcal{H}/\partial B$ and widens the value gap $V(n') - V((1-\varphi)n')$ in (14). In the calibrated model, τ rises roughly sixfold from its ergodic mean of 0.02 to about 0.13 as the economy approaches the constrained region. This sharp increase in the cost of deposits near the kink acts as an endogenous "guardrail": the closer the economy drifts toward the constrained region, the more aggressively the planner's wedge pushes back by penalizing leverage, creating a restoring force that is absent in the CE.

That said, the guardrail does not provide absolute protection. Sufficiently large or repeated distress events can still drive the planner's economy into the constrained region—though the constraint binds only about 0.2% of the time under the planner's policy, against 18% in the CE. But when it does bind, the starting point matters enormously. In the CE, the economy enters the constrained zone from a low equity base ($N \approx 0.009$), deep in the steep portion of the saddle path where each unit of equity loss triggers further deleveraging. The planner's economy enters the same zone from a much higher base ($N \approx 0.026$), close to the kink's boundary rather than deep within it, enabling a faster exit. The constraint binds *briefly and mildly* rather than *persistently and severely*: the planner converts what are prolonged credit crunches in the CE into transient episodes that resolve within a few periods.

The magnitude of τ reflects this geometric logic. The wedge is sizable: its ergodic mean is roughly half the CE deposit spread $R^B - 1 \approx 0.045$, and near the kink it climbs to about three times that spread, because the externality it corrects is severe: in the CE, the leverage constraint binds frequently, amplifying every distress event into a prolonged contraction. The standard deviation of τ (Table 5) is large relative to its mean, reflecting the highly nonlinear nature of the hazard function: the marginal value of equity as a crisis buffer rises sharply as the capital ratio declines toward the steep region of the sigmoid (6).

In a decentralized implementation, a state-contingent tax $\tau(X)$ on deposit issuance, rebated lump-sum, raises the effective cost of deposits from $R^B - 1$ to $R^B - 1 + \tau$ and induces banks to substitute toward equity financing. This corrects the externality on the *lending* margin. It does not, however, replicate the planner on its own: as Section 3.7 establishes, the planner's marginal value of equity carries the larger wedge $\nu = \tau L/N$, so fully decentralizing its equity accumulation also requires rewarding retention—the role played by the positive slope of the leverage rule in Section 5.5, which makes a binding cap's shadow value $\lambda(1+\phi_1)$ stand in for ν. Because the wedge varies with the aggregate state $X=(N,A)$, a fixed capital surcharge would only approximate it; the time variation in τ benchmarks the state-contingency that countercyclical capital buffers should exhibit.

For TFP shocks, however, the planner faces the same technology constraint. Productivity shocks do not trigger failures, so the fragility externality is not activated and the planner cannot improve upon the market outcome. This clarifies the scope of macroprudential policy: it is effective where the market failure lies (the systemic undervaluation of equity as a buffer against financial distress) and offers no advantage where shocks originate in the real sector.

5.3 Leverage limits vs. Pigouvian taxation

The Pigouvian tax $\tau(X)$ is a state-contingent instrument that is informationally demanding: it requires the regulator to observe the aggregate state and compute the marginal externality in real time. A simpler alternative is to tighten the leverage constraint directly by lowering ϕ_1. This exercise isolates what a blunt, time-invariant capital requirement can and cannot achieve.

Figure 10 compares the conditional saddle paths under three regimes: the baseline CE ($\phi_1=1.5$), a tighter leverage constraint ($\phi_1=0.8$), and the SPP. Tightening the constraint shifts both saddle paths rightward, raising the conditional steady states and moving the economy away from the kink—qualitatively similar to the Pigouvian tax. However, the mechanism differs in two important respects.

First, the tight constraint operates by *compressing credit across all states*. By capping deposits at a lower multiple of equity, it forces banks to shrink their balance sheets even when the economy is well-capitalized and the fragility externality is negligible. This flattens the saddle paths, reducing the across-saddle gap $\Delta C(N)$ and dampening output in normal times. The SPP, by contrast, achieves its equity targets through a state-contingent wedge that is small when the economy is healthy and large only near the kink. The planner's saddle paths retain their slope, so credit intermediation is unimpaired in good states, while selectively steepening the penalty for leverage when it matters.

Second, the tight constraint cannot fully replicate the planner's allocation because it addresses the *symptom* (excessive leverage) rather than the *cause* (the uninternalized hazard externality). The constraint binds roughly a quarter of the time under the tight policy, versus only 0.2% under the planner. When it binds, it prevents deleveraging spirals—but it also prevents the economy from reaching the efficient level of intermediation. The planner achieves both far lower binding frequency *and* higher mean output, revealing the efficiency cost of the blunt instrument.

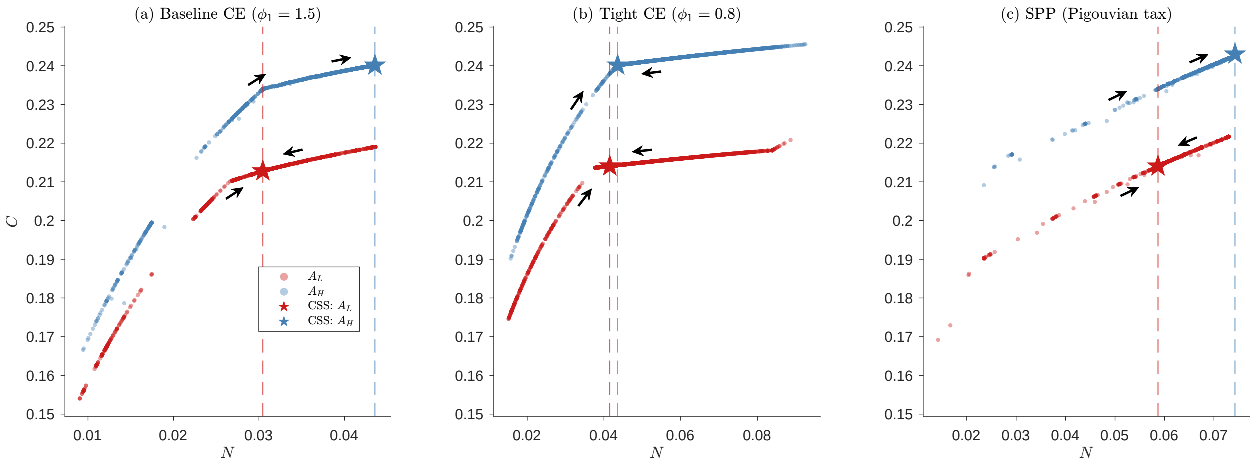

Figure 10: Conditional saddle paths: baseline CE, tight leverage constraint, and SPP.

Notes: Saddle paths under frozen TFP regimes. Stars mark conditional steady states. Panel (a): baseline CE ($\phi_1 = 1.5$). Panel (b): tight CE ($\phi_1 = 0.8$). Panel (c): SPP. All panels share a common vertical axis for direct comparison (the horizontal ranges adapt to each economy's equity support). The tight constraint shifts saddle paths rightward but also compresses them vertically, reflecting reduced credit intermediation.

The comparison highlights a general principle: leverage limits and Pigouvian taxes are complementary rather than substitutable instruments. Leverage limits provide a robust floor that does not require real-time state observation, but at the cost of compressing credit intermediation uniformly. The Pigouvian wedge targets the externality precisely but requires the regulator to track the aggregate state. In practice, a well-designed macroprudential framework would combine a time-invariant leverage floor (a relaxed version of the tight constraint) with a countercyclical buffer (approximating the state-contingent wedge) that activates as the system approaches the fragile region. The remainder of this section formalizes both ideas: Section 5.4 traces the policy frontier as ϕ_1 varies, and Section 5.5 derives a macroprudential rule from the planner's solution and interprets its structure.

5.4 Comparative statics of leverage policy

Before turning to state-dependent instruments, we use a fixed leverage limit as a policy-frontier experiment: it asks how far a simple, time-invariant rule can move the economy toward the planner's allocation and what efficiency cost it pays relative to the state-contingent Pigouvian benchmark. Table 6 reports ergodic outcomes as ϕ_1 varies from 0.05 (very tight) to 2.00 (loose), holding all other parameters at their baseline values. Three regularities emerge.

First, tightening the constraint monotonically reduces fragility. The mean hazard rate falls from roughly 12% at $\phi_1 = 2.00$ to about 0.5% at the tightest $\phi_1 = 0.05$, and crisis frequency declines proportionally. The mechanism is direct: a lower ϕ_1 caps deposits at a tighter multiple of equity, raising the *ergodic* capital ratio N/L (reflected here in the falling mean hazard $\bar{\mathcal{H}}$) and pushing the economy into the flat region of the sigmoid hazard function (6). (The *deterministic* steady-state ratio, by contrast, is invariant to ϕ_1; it is what the ergodic ratio rises above as the cap tightens.)

Second, the binding frequency of the constraint is highly nonlinear in ϕ_1. At $\phi_1 = 0.50$ the constraint binds roughly two-thirds of the time, acting as a near-permanent cap on intermediation. At the baseline $\phi_1 = 1.50$ it binds about 18% of the time, only during distress episodes. This nonlinearity implies that a regulator setting a constant leverage limit faces a steep trade-off: the binding frequency is nearly flat down to $\phi_1 = 1.0$ (about 18%) but rises steeply once the cap falls below that, to two-thirds of all periods at $\phi_1 = 0.5$ and essentially every period for $\phi_1 \leq 0.3$ (Table 6).

Third, no constant cap matches the planner on welfare. Measured inclusive of the transition (eq. (18)), the planner weakly dominates every constant cap that improves on the competitive equilibrium, since any such cap's allocation is itself available to the planner. The cap frontier traces an inverted U: tightening helps until the accumulation cost overtakes the fragility benefit. Among the caps we solve exactly, the transition-inclusive gain peaks near $\phi_1 \approx 0.1$ at about $+1\%$ (with $\phi_1 = 0.10$ delivering $+1.03\%$ and the tighter $\phi_1 = 0.05$ already turning down to $+0.97\%$), well short of the planner's $+1.34\%$. The residual is the cost of bluntness: a constant ϕ_1 compresses credit *uniformly*, forcing needless equity accumulation even in safe states where intermediation is valuable, whereas the planner targets the externality *selectively*, building the buffer precisely where fragility is high while leaving intermediation unimpaired when the system is safely capitalized. This gap between the precise and the blunt instrument motivates the search for a state-dependent rule.

Unlike the planner, whose price signal leaves the constraint binding only 0.2% of the time, a constant tighter leverage ratio achieves stability through near-permanent rationing: a tight fixed cap ($\phi_1 \leq 0.3$) binds essentially every period and even the moderate $\phi_1 = 0.5$ binds two-thirds of the time (Table 6). The gap is the efficiency cost of correcting a state-dependent externality with a state-independent quantity limit—the cap represses credit even in states where the system is safely capitalized and fragility is low. This motivates the two-instrument rule of the next subsection, whose repositioned intercept and slope deliver the planner-like capital ratio without conscripting the constraint into permanent service.

Table 6: Comparative Statics: Leverage Slope ϕ_1

ϕ_1	$\bar{\mathcal{H}}$	Binding (%)	$\sigma(Y)$	$\bar{Y}$
0.05	0.0051	100.0	0.0200	0.3340
0.10	0.0065	100.0	0.0206	0.3341
0.14	0.0079	100.0	0.0210	0.3341
0.20	0.0103	100.0	0.0219	0.3337
0.30	0.0151	100.0	0.0228	0.3333
0.50	0.0275	66.0	0.0251	0.3316
1.00	0.0569	18.2	0.0289	0.3283
1.50[†]	0.0882	17.5	0.0310	0.3257
2.00	0.1190	20.0	0.0316	0.3236
Planner	0.0088	0.2	0.0201	0.3352

Notes: [†]Baseline calibration. $\bar{\mathcal{H}}$ is the mean ergodic hazard, which falls as ϕ_1 tightens (the ergodic capital ratio rises even though the deterministic steady-state ratio, 0.4975, is invariant to ϕ_1). Binding: fraction of periods with an active leverage constraint. $\sigma(Y)$: output volatility; $\bar{Y}$: mean output. Every row, including the baseline, is re-solved on the same solver as the planner, so the baseline moments differ marginally from Table 4. Welfare is reported separately (Table 7 and Section 5.4): measured inclusive of the transition (eq. (18)), the constant-cap gain is an inverted U in ϕ_1, peaking near $\phi_1 \approx 0.1$ at about $+1\%$ and lying below the planner's $+1.34\%$ for every cap the planner could itself adopt.

5.5 A macroprudential rule from the planner's solution

We now turn from the policy-frontier experiment to implementation. The question is not whether a fixed rule improves on laissez-faire, but how closely a practical leverage rule can approximate the planner's constrained-efficient correction. We implement the planner through a two-instrument leverage cap, $B \leq \phi_0 + \phi_1 N$, whose coefficients play distinct roles: the slope ϕ_1 prices the fragility externality (it is the Pigouvian wedge after the constraint's shadow value), while the intercept ϕ_0 sets the leverage level. A regulator who conditions on both can separate credit growth that erodes the sector-wide capital ratio from credit growth backed by strong fundamentals—a distinction that aggregate credit-gap rules such as the Basel III CCyB (Basel Committee on Banking Supervision, 2010) collapse into a single credit-growth response. The closest antecedent is the optimized macroprudential Taylor rule of Bianchi and Mendoza (2018), which targets a Pigouvian tax on debt in a small open economy. Our rule differs on two dimensions. The instrument is a leverage coefficient on bank equity rather than a debt tax, mapping directly to Basel-style capital requirements. The hazard function is calibrated empirically using the Jordà et al. (2017) crisis database. Separately from these differences, the global solution also matters: it estimates the rule over the nonlinear region around the leverage kink, where local approximations would understate the state dependence.

5.5.1 A two-instrument leverage rule

We implement the planner through a leverage rule of the same form the competitive bank already faces, $B \leq \phi_0 + \phi_1 N$, but treat *both* coefficients as policy instruments. They map cleanly onto the planner's two corrections. The competitive bank's marginal value of equity is $J_1 = R^B + \lambda(1+\phi_1)$, where λ is the multiplier on the cap. The term $\lambda(1+\phi_1)$ is the value of equity from relaxing tomorrow's constraint: one unit of retained equity raises next period's deposit capacity by ϕ_1. The *slope* ϕ_1 is therefore the instrument that rewards capitalization. The *intercept* ϕ_0 shifts the cap up or down uniformly and carries no such incentive, since $\partial(\text{cap})/\partial N = \phi_1$ is independent of ϕ_0.

The planner's marginal value of equity is $J_1^{\text{SPP}} = R^B + \nu$, with $\nu = \tau L/N$ the equity wedge of Section 3.7 (the planner's own cap barely binds, so its $\lambda \approx 0$). Equating the two fixes the role of each instrument:

$$\underbrace{\lambda\,(1+\phi_1)}_{\text{constraint's shadow value}} \quad = \quad \underbrace{\nu}_{\text{Pigouvian equity wedge}} \quad . \tag{19}$$

After the multiplier, ϕ_1 *is* the Pigouvian price of the fragility externality: it is set so that a binding leverage cap reproduces the social reward to capital. Given ϕ_1 (and the λ it induces), the intercept is the residual that positions the cap at the planner's leverage level, $\phi_0 = B^{\text{SPP}} - \phi_1 N$. The single-instrument rule that fixes ϕ_0 and reads all variation into the slope conflates these two roles.

5.5.2 The planner's leverage is nearly linear—but its slope cannot be the rule

Freeing the intercept reveals that the planner's deposit policy is almost exactly linear in equity. Regressing B^{SPP} on N by TFP regime gives

$$B^{\text{SPP}}(N, A) \;\approx\; \phi_0(A) + \phi_1(A)\,N, \qquad R^2 = 0.99, \tag{20}$$

with $R^2 = 0.98$ in recessions and 0.998 in expansions, against $R^2 = 0.87$ for the constrained fit that holds ϕ_0 at its baseline value. The planner runs a clean linear leverage schedule; the constrained specification missed it only by forcing the wrong intercept. The state dependence sits almost entirely in the intercept, with ϕ_0 rising from 0.059 in recessions to 0.068 in expansions (a looser baseline allowance when fundamentals are strong) while the slope is essentially acyclical.

That slope, however, is *negative* in every specification ($\phi_1 \approx -0.39$ pooled, ≈ -0.69 within regime, and -0.83 in the output-conditioned fit of Table 7, Panel A): the planner *deleverages* as it accumulates, substituting equity for deposits as its buffer grows—the descriptive footprint of the retention wedge ν. The rule's slope is a different object. A negative slope would invert (19), making $\lambda(1+\phi_1)$ *small* and so weakening rather than supplying the capitalization incentive. The implementable rule must therefore price the externality through a *positive* slope, not trace the planner's realized leverage.

5.5.3 The optimal implementable rule

The welfare-maximizing two-instrument rule therefore sets $\phi_1 > 0$ to supply the capitalization incentive, with ϕ_0 pinning the level. Searching over (ϕ_0, ϕ_1) yields a broad optimum: any $\phi_1 \in [0.14, 0.30]$, with ϕ_0 adjusted to hold the leverage level, recovers about 97% of the planner's constrained optimum (+1.30% against the planner's +1.34%) and nearly reproduces the planner's ergodic capital ratio and crisis frequency ($N/L = 0.79$ vs. the planner's 0.78; crisis frequency 0.7% vs. 0.9%). Table 7 summarizes the planner's fitted policy (Panel A), the welfare ladder (Panel B), and the implementable rule's enforced coefficients and outcomes (Panel C).

The optimum is a *plateau* rather than a sharply state-contingent schedule. Because the planner's leverage is nearly acyclical, a tight, roughly constant cap with $\phi_1 > 0$ already captures almost all of the available gain: the slope's job is to price the externality, not to track the cycle. The genuine state dependence in the planner's behavior (running high leverage when capital is scarce and deleveraging as it rebuilds) operates through equity *retention*, which a leverage instrument cannot reproduce. The residual between the optimal rule and the planner's constrained optimum, about 0.04 percentage points, is precisely the uncorrected equity wedge ν.

Written out, the rule takes a familiar Taylor form in two *observable* aggregates—bank capital and output:

$$B \;\leq\; \bar{\phi}_0 + \gamma_Y\left(\frac{Y}{\bar{Y}} - 1\right) + \phi_1\, N, \tag{21}$$

with a baseline allowance $\bar{\phi}_0$, an output term ($\gamma_Y > 0$) that loosens the level in expansions, and a capital term ($\phi_1 > 0$) that prices the externality. We condition the business-cycle margin on output rather than TFP because output is what a regulator observes, whereas productivity is a latent residual; output is in fact the better summary statistic here, raising the fit of the planner's deposit policy from $R^2 = 0.99$ to 0.998. The slope does the welfare work: making the level cyclical leaves the gain essentially unchanged (the recovery stays near 97% with or without it). The separation the rule embodies is therefore *informational* rather than an additional source of welfare: it tells the regulator to read a credit boom's source, tightening on leverage-driven expansion while accommodating output-driven growth. Because it reads bank capital and output as *separate* signals, the rule draws a distinction that the credit-to-GDP gap, which merges credit and output into a single ratio, cannot.

5.5.4 Reading the rule

Two features of the optimal rule deserve emphasis. First, its baseline is *tight*: the level ϕ_0 and slope ϕ_1 together sustain the planner's capital ratio of $N/L \approx 0.8$, roughly double the competitive economy's 0.4. The planner demands this not because individual banks are imprudent but because their collectively rational leverage produces a systemically fragile outcome; the rule's tightness is a measure of the externality's severity. Second, the rule conditions on two *observable* signals, bank capital and output, *separately*: the level is looser when output is high ($\gamma_Y > 0$), while the binding slope prices fragility according to the current capital ratio. This lets a regulator distinguish credit growth that erodes the sector-wide capital ratio, which warrants tightening, from output-driven growth, which does not. A single credit-to-GDP gap, which merges credit and output into one ratio, cannot make the distinction.

5.5.5 Limitations of the leverage instrument

While the linear rule achieves a high R^2, it is important to emphasize what it does *not* accomplish. The leverage constraint controls deposits, one of two margins the planner corrects, priced by the deposit wedge τ. The other is equity retention: the choice of n'. In the SPP, this margin carries the larger equity wedge $\nu = \tau\, L/N$, entering the envelope condition as $J_1^{\mathrm{SPP}} = R^B + \lambda(1+\phi_1) + \nu$ and raising the marginal value of equity directly. A binding leverage constraint provides equity retention only an *indirect* bonus, $\lambda(1+\phi_1)$, determined by the constraint's shadow value rather than by the externality itself. This bonus is nonetheless a powerful, if imperfect, substitute: by tuning ϕ_1 so that $\lambda(1+\phi_1)$ stands in for the equity wedge ν (19), the optimal rule recovers about 97% of the planner's welfare gain. The uncorrected sliver of ν is the residual few basis points. Closing it would require pairing the leverage rule with an instrument that rewards equity retention directly, such as a tax on payouts or a subsidy to retained earnings.

Two practical lessons follow from this "one instrument, two margins" structure. First,

Table 7: The planner's leverage policy and its decentralization

Panel A. Planner's deposit policy $B = \bar{\phi}_0 + \gamma_Y (Y/\bar{Y} - 1) + \phi_1 N$	
$\bar{\phi}_0$ (baseline level)	0.072
γ_Y (output, observable)	$+0.079$
ϕ_1 (capital slope)	-0.828
R^2	0.998
R^2 (TFP A instead of output)	0.990
R^2 (ϕ_0 fixed at 0.01)	0.87
Panel B. Welfare of the leverage rule (CEV vs. CE)	
Planner (constrained optimum)	$+1.34\%$
Optimal two-instrument rule ($\phi_1 > 0$)	$+1.30\%$
Best constant cap	$\approx +1.0\%$
Competitive equilibrium	0
Panel C. The implementable rule: enforced coefficients and outcomes	
ϕ_1 (capital slope)	0.14 (plateau over $[0.14, 0.30]$)
ϕ_0 (level: recession / expansion)	0.004 / 0.014 ($\gamma_Y > 0$)
ϕ_0 (constant-level variant)	0.01
Ergodic capital ratio N/L	0.78 (planner: 0.78)
Crisis frequency	0.7% (planner: 0.9%)
Constraint binding frequency	$\approx$100% (planner: 0.2%)

Notes: Panel A regresses the planner's deposits on equity and the output gap over the ergodic simulation ($T = 5{,}001$). Output Y is the observable business-cycle signal (TFP is a latent residual); conditioning on it raises the fit to $R^2 = 0.998$, and freeing the intercept ϕ_0 raises the fit from 0.87 to 0.99 even with TFP. The fitted capital slope is *negative*—the planner deleverages as it accumulates; the rule's slope, by contrast, is *positive*, set to price the externality via $\lambda(1 + \phi_1)$ rather than to trace the planner's realized leverage. Panel B reports the transition-inclusive consumption-equivalent welfare gain (eq. (18), which prices the cost of building equity from the mean of the CE's ergodic equity distribution) from imposing each policy on the competitive economy. The optimal rule is a plateau over $\phi_1 \in [0.14, 0.30]$, recovering $\approx 97\%$ of the planner's constrained optimum; the residual reflects the equity wedge ν, which a leverage instrument cannot price. The planner dominates every constant cap it could itself adopt (Section 5.4), so the best cap falls short of it. Panel C reports the coefficients of the optimal implementable rule in the notation of eq. (21)—the binding slope $\phi_1 = 0.14$ paired with a level that loosens in expansions ($\phi_0 = 0.004$ in recessions, 0.014 in expansions)—and its delivered outcomes; the constant-level variant ($\phi_0 = 0.01$) forgoes almost none of the welfare gain, which is why the cyclical level is informational rather than welfare-driving. The rule works *through* the binding constraint: its shadow value $\lambda(1 + \phi_1)$ supplies the capitalization incentive, so near-universal binding is the mechanism, not a malfunction.

a tight leverage cap with a *positive* slope is a remarkably effective instrument: it recovers nearly all of the planner's gain even though it acts only on the quantity of deposits, because a binding constraint manufactures the equity incentive endogenously through its shadow value. Second, what it cannot supply is the final increment—the part of the fragility externality that is a reward to *retained equity* rather than a tax on deposits. The rule is thus both a near-complete decentralization and a clean diagnosis of where a second instrument would have to act.

5.5.6 The value of making the rule common knowledge

Finally, we ask whether the rule's welfare content depends on banks *anticipating* its state-contingent structure, as opposed to merely facing its realized values. We compare two rational-expectations equilibria under the same enforced state-contingent cap schedule: one in which banks correctly condition on how the cap varies with (N, A), the other in which they treat the realized cap as an i.i.d. draw around its unconditional mean. We find that making the structure common knowledge is robustly welfare-improving but second-order: a consumption-equivalent gain of $+0.061\%$ (95% confidence interval $[+0.055\%, +0.068\%]$ across $n = 10$ simulation seeds). The gain operates through intermediation efficiency rather than financial stability: anticipating that the cap loosens in expansions, banks time their borrowing slightly better, while the mean failure probability is essentially unchanged across the two economies.

5.6 Anatomy of the SPP–CE capitalization gap

Proposition 1 establishes that the equity wedge ν is positive in every state, so the planner's marginal valuation of equity exceeds the market's; in the calibrated solution it accumulates more equity than the competitive equilibrium, $N^{\text{SPP}} > N^{\text{CE}}$. Because TFP shocks are exogenous and symmetric across agents, no individual bank's actions can alter them, so there is no externality to correct on that margin. The failure hazard, by contrast, is an endogenous aggregate outcome: each bank takes $\mathcal{H}$ as given, yet its leverage choice contributes to the sector-wide capital ratio that determines $\mathcal{H}$. The Pigouvian wedge $\tau > 0$ prices exactly this missing feedback. Consequently, the *direction* of the correction is unambiguous (more equity is warranted), and the operative question is *how much*. This section decomposes the quantitative magnitude of the SPP–CE gap into three structural channels and shows, via a counterfactual exercise, that the gap is highly sensitive to two primitive parameters: the severity of bank failure (φ) and the volatility of TFP shocks (σ_A).

5.6.1 Three-channel decomposition

The envelope conditions for the CE and SPP are:

$$J_1^{\mathrm{CE}}(N;X) = R^B + \lambda\,(1+\phi_1), \tag{22}$$

$$J_1^{\mathrm{SPP}}(N;X) = R^B + \lambda\,(1+\phi_1) + \nu, \qquad \nu = \tau\,\frac{L}{N}. \tag{23}$$

At each economy's respective ergodic distribution, the Euler equation pins down the average marginal value of equity. The gap $J_1^{\mathrm{SPP}} - J_1^{\mathrm{CE}}$ can be decomposed into three channels evaluated at the ergodic means—an *ergodic-accounting* decomposition across the two economies' *distinct* stationary distributions, not a same-state marginal comparison:

$$\underbrace{\mathbb{E}\left[J_1^{\mathrm{SPP}}\right] - \mathbb{E}\left[J_1^{\mathrm{CE}}\right]}_{\text{total gap}} = \underbrace{\Delta R^B}_{\text{deposit rate}} + \underbrace{\Delta[\lambda\,(1+\phi_1)]}_{\text{shadow value}} + \underbrace{\mathbb{E}[\nu]}_{\text{equity wedge}}, \tag{24}$$

where $\Delta R^B \equiv \mathbb{E}[R^{B,\mathrm{SPP}}] - \mathbb{E}[R^{B,\mathrm{CE}}]$ and $\Delta[\lambda(1+\phi_1)]$ is defined analogously. Although Proposition 1 guarantees $\nu > 0$, the other two channels are *negative* in equilibrium, partially offsetting the equity wedge:

- **[+] Equity wedge** ($\nu > 0$): The planner internalizes that higher equity reduces the failure hazard *and*, unlike deposits, expands the balance sheet against which fragility is measured, so the social value of equity carries the scaled wedge $\nu = \tau\, L/N$—a positive marginal value absent in the CE. This channel pushes the SPP toward *more* equity.

- **[−] Deposit rate gap** ($\Delta R^B < 0$): The SPP restricts deposits to reduce leverage. With lower B, the marginal deposit cost $\xi'(B) = \zeta_1 + 2\zeta_2 B = 2\zeta_2 B$ falls, so $R^B = 1 + \xi'(B)$ is lower for the SPP. A lower deposit rate reduces the return on equity, pushing the SPP toward *less* equity.

- **[−] Shadow value gap** ($\Delta[\lambda(1+\phi_1)] < 0$): The SPP's lower leverage means its constraint binds far less frequently (0.2% vs. 18% at baseline). The resulting lower average λ reduces the equity return, again pushing the SPP toward *less* equity.

Table 8 reports the decomposition at baseline calibration. The equity wedge ($\nu = 0.031$) is positive but *smaller* than the combined offset from the deposit rate and shadow value channels ($-0.027 - 0.108 = -0.135$). The net gap is negative (-0.105; the components are rounded, so they sum to -0.104): $\mathbb{E}[J_1^{\mathrm{SPP}}] < \mathbb{E}[J_1^{\mathrm{CE}}]$. This is *consistent* with Proposition 1:

the SPP holds more equity, and at higher N the marginal return to equity is lower due to diminishing returns from intermediation. The planner accumulates equity precisely *until* the combined return, inclusive of the Pigouvian bonus, falls to the household's required rate. The negative net gap reflects the fact that the SPP's equilibrium equity is high enough to exhaust the excess returns that the CE leaves on the table.

Table 8: Decomposition of the SPP–CE Return Gap

	Baseline ($\sigma_A = \pm 2\%$, $\varphi = 0.60$)	Counterfactual ($\sigma_A = \pm 5\%$, $\varphi = 0.20$)	Direction
Equity wedge $\mathbb{E}[\nu]$	$+0.031$	$+0.018$	[+] more N
Deposit rate gap ΔR^B	-0.027	-0.013	[$-$] less N
Shadow value gap $\Delta[\lambda(1+\phi_1)]$	-0.108	-0.018	[$-$] less N
Net gap	-0.105	-0.013	
$N^{\text{SPP}}/N^{\text{CE}} - 1$	$+95\%$	$+32\%$	

Notes: Ergodic means from the baseline ($\sigma_A = \pm 2\%$, $\varphi = 0.60$) and counterfactual ($\sigma_A = \pm 5\%$, $\varphi = 0.20$) calibrations. The equity wedge $\nu = \tau L/N$ inherits the approximately linear scaling of τ in φ; the shadow value gap depends on TFP volatility through the frequency with which the leverage constraint binds.

5.6.2 Counterfactual: high TFP volatility, low failure severity

To assess how much the SPP–CE gap depends on primitives, we re-solve the model under a counterfactual calibration that widens TFP volatility ($\sigma_A = \pm 5\%$ vs. baseline $\pm 2\%$) while reducing the bank failure severity ($\varphi = 0.20$ vs. baseline 0.60). Both changes shrink the SPP–CE gap, but through different channels. Lower φ shrinks the wedge *itself* (τ, and hence $\nu = \tau L/N$): bank failures are less costly, so the externality the planner corrects is smaller. Wider TFP shocks, meanwhile, lead the CE to accumulate a larger precautionary equity buffer, so its leverage constraint binds *less* often and the offsetting shadow-value gap compresses (as detailed below).

As the right column of Table 8 reports, the SPP–CE equity gap shrinks substantially in the counterfactual: from 95% at baseline to 32%. The equity wedge falls from $\nu = 0.031$ to 0.018 as bank failures become less costly, so the planner's additional equity retention is much more modest.

The mechanism operates through the interaction of the three channels. In the counterfactual, the underlying Pigouvian object falls roughly in proportion to φ (which drops from 0.60 to 0.20), since for small φ, $V(n') - V((1-\varphi)n') \approx \varphi \cdot V'(n') \cdot n'$; the equity wedge $\nu = \tau L/N$ inherits this scaling, falling from 0.031 to 0.018. Meanwhile, the CE leverage constraint binds less frequently (9% vs. 18%) despite wider TFP, because the CE itself accumulates more equity as a buffer against larger shocks. The planner's constraint binds only 0.3% of the time (vs. 0.2% at baseline), and the shadow value gap between the two economies narrows sharply (from -0.108 to -0.018). Finally, the deposit rate gap shrinks because the planner, facing a smaller externality, restricts deposits less aggressively.

In the limit $\varphi \to 0$, bank failures become costless, $\tau \to 0$ and hence $\nu \to 0$, and the SPP collapses to the CE. The equity wedge is the single structural force separating the two economies, and its magnitude is governed by the severity of bank failure.

Because the equity wedge is strictly positive, the *direction* of the correction is never in doubt: the planner's marginal valuation of equity exceeds the market's at every state, and in the calibrated economy it holds more equity than the market provides. The economically important question is *how much more*. The three-channel decomposition (24) provides a structural answer. The case for large macroprudential capital buffers is strongest when bank failures are costly and TFP volatility is moderate—the configuration in which the equity wedge is largest relative to the offsetting deposit-rate and shadow-value channels. When failure costs are low (e.g., due to effective resolution regimes or deposit insurance), the wedge shrinks and the two offsetting channels largely neutralize it, so the optimal buffer narrows from 95% to 32% of CE equity. The decomposition thus serves as a diagnostic for calibrating countercyclical capital buffers: the buffer should be large enough to close the Pigouvian gap but not so large as to overshoot the externality it corrects.

6 Conclusion

This paper develops a general equilibrium model that nests the credit exposure and fragility buffer channels within a single framework. Doing so reveals a capitalization dilemma: bank equity amplifies TFP shocks through the credit exposure channel but mitigates financial shocks through the fragility buffer channel. Because both channels operate through the same state variable, a social planner who internalizes the fragility externality cuts crisis frequency roughly tenfold and prevents distress events from becoming credit crunches, while leaving the economy more exposed to negative productivity shocks.

From a policy perspective, we show that the planner's allocation is closely approximated by a simple Taylor-rule-type leverage rule, $B \leq \phi_0 + \phi_1 N$. The planner's own deposit policy is nearly exactly linear in equity ($R^2 = 0.99$ once the intercept is freed), but its slope is *negative* (it deleverages as it rebuilds capital), so it cannot be handed to a regulator directly: a negative slope would weaken the very incentive to capitalize. The implementable rule instead uses a *positive* slope, which after the leverage constraint's shadow value is the Pigouvian price of the fragility externality, paired with an intercept that sets the level. So calibrated, a tight leverage cap recovers about 97% of the planner's welfare gain; the small residual is the part of the externality that rewards retained equity rather than taxing deposits, which a leverage instrument cannot reach. Because the rule conditions on bank capital and output, both observable, it distinguishes credit growth that erodes the sector-wide capital ratio from output-driven growth, a distinction lost in composite indicators such as the credit-to-GDP gap used in the Basel III CCyB framework.

Optimal policy does not eliminate the trade-off between resilience and macroeconomic volatility; it operates on the frontier between them. The macroprudential rule we derive offers a practical way to move along that frontier, capturing most of the stability gains from higher capital while limiting its cost in normal times.

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